AN EXACT FORMULA FOR OPTIMAL AXIS-PARALLEL PACKING OF EIGHT SQUARES

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Abstract. We determine the smallest square container for eight axis-parallel squares of prescribed side lengths. The optimal side length is the minimum of the costs of 23 explicit packing strategies, and each cost is a maximum of sums of side lengths. No minimum of fewer convex functions represents it. Three strategies suffice for seven squares; the jump to 23 for eight squares witnesses the beginning of a combinatorial explosion. We prove the formula by combining the constructions with geometric lower bounds derived from disjoint coordinate projections, cornering arguments, and a weighted area comparison.

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1. Introduction

Container problems ask for the smallest container of a given shape that holds prescribed convex bodies with pairwise disjoint interiors. Packing squares in a square already leads to difficult questions. Moon and Moser [MM67] showed that axis-parallel squares of total area at most $1/2$ always fit in a unit square, and that $1/2$ is sharp. Erdős and Graham [EG75] showed that unit squares, suitably tilted, can be packed into a square of large side a leaving uncovered area $O(a^{7/11})$. Friedman's survey [Fri09] describes the smallest squares known to contain n unit squares when rotations are allowed, and the cases for which optimality has been proved. For example, Stromquist [Str03] proved that ten unit squares require a square of side $3 + 1/\sqrt{2}$. A computer-assisted proof of

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the optimality of Walter Trump's 1979 packing of eleven unit squares has recently been announced [Que26b], together with a Lean formalization [Que26a].

For items of fixed orientation, Murata, Fujiyoshi, Nakatake, and Kajitani [MFNK96] encode placements of rectangles by sequence-pairs, and Fekete and Schepers [FS04] characterize feasible orthogonal packings in every dimension by packing classes. Encodings of this kind underlie exact search methods for individual instances. For example, Huang and Korf [HK13] find all minimum-area enclosing rectangles for a given set of rectangles, and extend the known solutions of the benchmark of consecutive squares $1, 2, \dots, n$ to $n = 32$. When the number of items is part of the input, deciding whether prescribed squares fit without rotations into a prescribed square is strongly NP-complete [LTW$^+$90].

Here the side lengths are arbitrary, the orientations are fixed, and the goal is an exact formula valid for every ordered side vector. Given a nonincreasing sequence $s = (s_0, s_1, \dots)$ of nonnegative numbers, let $L^*(s)$ be the least side of a square that contains axis-parallel squares $Q_0, Q_1, \dots$ of sides $s_0, s_1, \dots$ with pairwise disjoint interiors. Sides of length zero signify absent squares. The value $L^*(s)$ is finite exactly when $s \in \ell_2$, and we write $\ell_2^{\downarrow}$ for the cone of nonincreasing nonnegative sequences in ℓ_2. We identify a finite nonincreasing nonnegative tuple with the sequence in $\ell_2^{\downarrow}$ obtained by appending zeros. For $n \geq 1$ and $s \in \ell_2^{\downarrow}$, we write $L_n^*(s) = L^*(s_0, \dots, s_{n-1}, 0, 0, \dots)$ for the optimum for the n largest squares. In [Bad26a] we gave exact formulas for L_n^* when $n \leq 7$, together with methods for proving lower bounds. In particular, for every $s \in \ell_2^{\downarrow}$,

$$L_6^*(s) = \max(s_0 + s_1,\ s_2 + s_3 + s_4,\ s_1 + s_4 + s_5), \tag{1.1}$$

$$L_7^*(s) = \max\big(L_6^*(s), \min\{\max(s_3 + s_4 + s_5 + s_6,\ s_1 + s_2 + s_5),$$
$$\max(s_0 + s_5 + s_6,\ s_1 + s_3 + s_6),\ s_1 + s_4 + s_5 + s_6\}\big). \tag{1.2}$$

A strategy for n squares prescribes, for each pair of squares, at least one axis on which they are separated and their order along each such axis. Its cost is the greatest side sum of a set of squares that the strategy orders linearly along one axis, and $L_n^*(s)$ is the minimum of the costs over all strategies. A single strategy suffices for $n = 6$. For $n = 7$, three strategies suffice; we call them top, center, and right, after the position of Q_6, and their costs are the three entries of the minimum in (1.2), each combined with L_6^* by a maximum. For eight squares we prove the following.

Theorem 1.1. *For every $s \in \ell_2^{\downarrow}$,*

$$L_8^*(s) = \max\big(L_7^*(s),\ \min\{\alpha(s), \beta(s), \dots, \psi(s)\}\big) = \min_{\omega} \max\big(L_6^*(s), \omega(s)\big),$$

where $\alpha, \beta, \dots, \psi$ are the twenty-three maxima of side sums in Table 1.1 and ω ranges over them. For each ω, the function $\ell_\omega = \max(L_6^*, \omega)$ is the cost of the strategy σ_ω defined by the coordinates in the row of ω in Table 3.1.

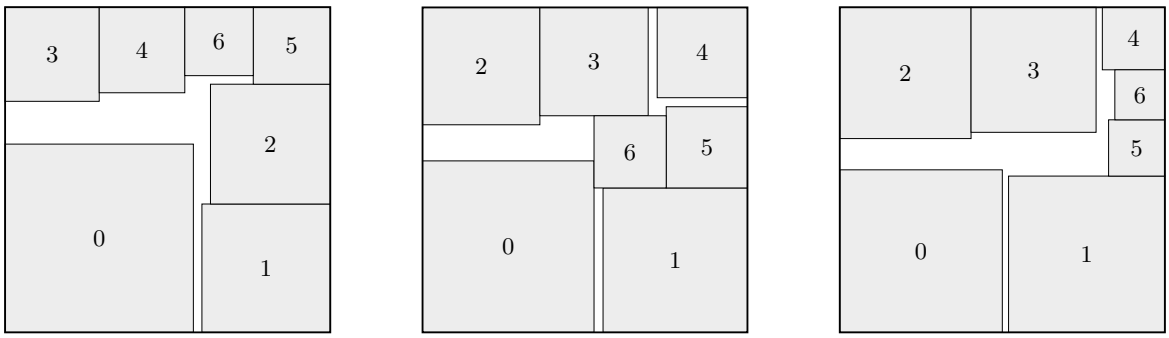
Moreover, if $h_1, \dots, h_m$ are real-valued convex functions on $\ell_2^{\downarrow}$ with $L_8^*(s) = \min_k h_k(s)$ for every $s \in \ell_2^{\downarrow}$, then $m \geq 23$. In particular, every complete family of strategies for eight squares has at least twenty-three members.

TABLE 1.1. The twenty-three maxima of side sums in Theorem 1.1.

$$\alpha = \max(s_0 + s_4 + s_7,\ s_0 + s_5 + s_6,\ s_1 + s_2 + s_7,\ s_1 + s_3 + s_5)$$
$$\beta = \max(s_0 + s_3 + s_7,\ s_0 + s_5 + s_6,\ s_1 + s_3 + s_6,\ s_2 + s_5 + s_6 + s_7)$$
$$\gamma = \max(s_0 + s_3 + s_7,\ s_0 + s_5 + s_6,\ s_1 + s_2 + s_6)$$
$$\delta = \max(s_0 + s_4 + s_5,\ s_1 + s_2 + s_7,\ s_1 + s_3 + s_6)$$
$$\varepsilon = \max(s_0 + s_5 + s_6,\ s_1 + s_3 + s_4,\ s_2 + s_4 + s_6 + s_7)$$
$$\zeta = \max(s_0 + s_6 + s_7,\ s_1 + s_2 + s_4,\ s_3 + s_4 + s_5 + s_6)$$
$$\eta = \max(s_0 + s_3 + s_6,\ s_1 + s_2 + s_7)$$
$$\theta = \max(s_0 + s_2 + s_7,\ s_0 + s_5 + s_6,\ s_1 + s_3 + s_6)$$
$$\iota = \max(s_0 + s_4 + s_7,\ s_0 + s_5 + s_6,\ s_1 + s_3 + s_5,\ s_2 + s_5 + s_6 + s_7)$$
$$\kappa = \max(s_0 + s_5 + s_6,\ s_1 + s_2 + s_4,\ s_3 + s_4 + s_6 + s_7)$$
$$\lambda = \max(s_0 + s_4 + s_5,\ s_1 + s_3 + s_6,\ s_2 + s_5 + s_6 + s_7)$$
$$\mu = \max(s_0 + s_5 + s_6,\ s_1 + s_3 + s_5,\ s_1 + s_4 + s_6 + s_7)$$
$$\nu = \max(s_0 + s_6 + s_7,\ s_1 + s_3 + s_7,\ s_1 + s_4 + s_5 + s_6)$$
$$\xi = \max(s_1 + s_3 + s_5,\ s_1 + s_3 + s_6 + s_7,\ s_2 + s_4 + s_5 + s_6)$$
$$o = \max(s_0 + s_4 + s_7,\ s_1 + s_2 + s_5,\ s_3 + s_4 + s_5 + s_6)$$
$$\pi = \max(s_0 + s_5 + s_6,\ s_1 + s_3 + s_6 + s_7)$$
$$\rho = \max(s_0 + s_5 + s_6,\ s_1 + s_3 + s_6,\ s_1 + s_4 + s_5 + s_7)$$
$$\sigma = \max(s_1 + s_4 + s_6 + s_7,\ s_2 + s_3 + s_5 + s_6)$$
$$\tau = \max(s_1 + s_4 + s_5 + s_6,\ s_2 + s_3 + s_6 + s_7)$$
$$\upsilon = \max(s_1 + s_5 + s_6 + s_7,\ s_2 + s_3 + s_4 + s_6)$$
$$\varphi = \max(s_1 + s_2 + s_5,\ s_1 + s_2 + s_6 + s_7,\ s_3 + s_4 + s_5 + s_6)$$
$$\chi = \max(s_0 + s_5 + s_6 + s_7,\ s_1 + s_3 + s_6)$$
$$\psi = s_1 + s_4 + s_5 + s_6 + s_7$$

For each expression ω in Table 1.1, the coordinates in the row of ω in Table 3.1 give a packing in every square of side at least the strategy cost ℓ_ω; Figure 3.1 illustrates them. This proves the upper bound in the formula of Theorem 1.1. For the lower bound we prove eight new inequalities $L_8^*(s) \geq b_i(s)$, $i = 1, \ldots, 8$, expressed using minima and maxima of side sums of $s_0, \ldots, s_7$, and use one more, a column bound from [Bad26a]. Most of the new bounds follow from the chain and cornering arguments of [Bad26a], and one from a weighted comparison of areas. A comparison of side sums then shows that the maximum of L_7^*, the column bound, and $b_1, \ldots, b_8$ is at least the right-hand side of the formula. This step compares explicit formulas and requires no classification of optimal packings. The bound $m \geq 23$ in Theorem 1.1 follows from twenty-three side vectors p_ω, one for each ω, at which L_8^* fails to be subadditive for every pair of distinct vectors:

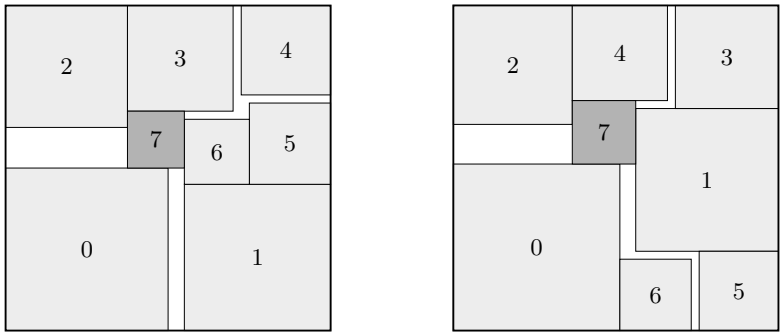
$$L_8^*(p_\omega + p_{\omega'}) > L_8^*(p_\omega) + L_8^*(p_{\omega'}) \quad \text{for all } \omega \neq \omega'.$$



(a) top, $L = 38$
$s = (22, 15, 14, 11, 10, 9, 8)$

(b) center, $L = 36$
$s = (19, 16, 13, 12, 10, 9, 8)$

(c) right, $L = 52$
$s = (26, 25, 21, 20, 10, 9, 8)$



(d) β, $L = 40$
$s = (20, 18, 15, 13, 11, 10, 8, 7)$

(e) α, $L = 41$
$s = (21, 18, 15, 13, 12, 10, 9, 8)$

FIGURE 1.1. Top row: packings of seven squares following top, center, and right of [Bad26a, Table 5.1], each with side $L_7^*(s)$. Bottom row: packings of eight squares following σ_β and σ_α, each with side $L_8^*(s)$; Q_7 is shaded.

An optimal strategy for eight squares need not restrict to an optimal strategy for the seven largest squares. Deleting Q_7 from σ_α leaves a strategy for $Q_0, \ldots, Q_6$ with cost $\max(L_6^*,\ s_0 + s_5 + s_6,\ s_1 + s_3 + s_5)$. At the side vector $s = (21, 18, 15, 13, 12, 10, 9, 8)$ of Figure 1.1(e), this cost is $41 = L_8^*(s)$, whereas $L_7^*(s) = 40$: the indices $5, 1, 3$ form a vertical chain of σ_α with side sum 41. In this sense the passage from seven to eight squares is nonlocal: an optimal arrangement of eight squares need not extend an optimal arrangement of the seven largest. By contrast, deleting Q_7 from σ_β leaves the arrangement center (Figure 1.1(b),(d)). Section 3 shows that the restrictions of thirteen of the twenty-three strategies have cost functions different from all three seven-square cost functions.

By [Bad26a, Theorem 6.2] and Theorem 1.1, the least sizes of complete families of strategies for six, seven, and eight squares are 1, 3, and 23. The successive ratios are 3 and 23/3, and the second exceeds the first by the factor 23/9. The lower-bound proofs grow as well: the proof for seven squares in [Bad26a] combines L_6^* with three inequalities, and the proof here combines L_7^* with eight new inequalities and the column bound of [Bad26a]. We do not know the least size for nine squares. If the ratio stayed at 23/3, the least size

would be about 176. If the ratio grew again by the factor 23/9, it would be about 19.6, giving about 450. Both estimates are speculative, but they suggest that an exact formula for L_9^* of the same type would be a minimum of hundreds of maxima of side sums.

For general n, a complete family of $m(n)$ strategies decides whether n prescribed squares fit into a prescribed square by evaluating $m(n)$ strategy costs, each the largest side sum of a chain in a pair of partial orders. Each strategy can be recorded by $O(n^2)$ order relations, and its cost can be computed in polynomial time: for either partial order, with vertex weights s_i, the largest side sum of a chain is the weight of a longest path in its directed graph, computed by the usual recurrence over predecessors. If $m(n)$ were bounded by a polynomial in n, these families would serve as advice of polynomial size for the strongly NP-complete problem of [LTW$^+$90]. Then NP $\subseteq$ P/poly, and the polynomial hierarchy would collapse to its second level by the Karp–Lipton theorem [KL82]. Thus, unless this collapse occurs, the least size of a complete family of strategies for n squares is not bounded above by any polynomial in n. In the other direction, the sequence-pair families of Silvanus and Vygen [SV20] give complete families of size $O\big(n!\,((11+5\sqrt{5})/2)^n/n^6\big)$; see [Bad26a] for a comparison of their notion of completeness with ours.

The rest of the paper is organized as follows. Section 2 collects the definitions and results of [Bad26a] that we use and proves two refinements of them. Section 3 gives the twenty-three strategies and their constructions. Section 4 proves the eight new lower bounds. Section 5 combines them with the column bound and L_7^* to prove the matching lower bound by comparing side sums, completes the proof of Theorem 1.1, and proves the bound $m \geq 23$. The code [Bad26b] verifies the constructions in Table 3.1, the packings in the figures, the comparisons following (3.2), and the computations in Section 5.2.

2. Preliminaries

We recall the conventions and geometric tools of [Bad26a] without proof, and prove a weighted-chain comparison and a refinement of its covering-chain lemma.

2.1. **Packings and strategies.** For each positive side s_i, write $Q_i = [x_i, x_i+s_i]\times[y_i, y_i+s_i]$. These squares form a packing in $[0, L]^2$ exactly when

$$0 \leq x_i, y_i \leq L - s_i \tag{2.1}$$

and, for each pair $i \neq j$, at least one of

$$x_i + s_i \leq x_j, \quad x_j + s_j \leq x_i, \quad y_i + s_i \leq y_j, \quad y_j + s_j \leq y_i. \tag{2.2}$$

Definition 2.1 (Strategies and complete families, [Bad26a, Definition 2.3])**.** A *strategy* for n squares is a pair $\sigma = (\prec_{\mathrm{h}}, \prec_{\mathrm{v}})$ of strict partial orders on $\{0, \dots, n-1\}$ such that every pair of distinct indices is comparable in at least one order. The relations prescribe horizontal and vertical separation. A *chain* of σ is a nonempty set of indices totally ordered by either relation. Write $\mathcal{C}(\sigma)$ for the set of these chains and define the *cost* of σ at a vector $v \in [0,\infty)^n$ by $\ell_\sigma(v) = \max_{I\in\mathcal{C}(\sigma)} \sum_{i\in I} v_i$. For $s \in \ell_2^{\downarrow}$, we write $\ell_\sigma(s)$ for its value at $(s_0, \dots, s_{n-1})$. A family $\mathcal{S}$ is *complete* if $L_n^*(s) = \min_{\sigma\in\mathcal{S}} \ell_\sigma(s)$ for every $s \in \ell_2^{\downarrow}$.

Lemma 2.2 (Strategy representation, [Bad26a, Proposition 2.5])**.** *For every $n \geq 1$ and every $s \in \ell_2^{\downarrow}$,*

$$L_n^*(s) = \min_{\sigma \in \mathfrak{S}_n} \ell_\sigma(s), \tag{2.3}$$

where $\mathfrak{S}_n$ is the finite set of all strategies. In particular, $L_n^*(s)$ is a continuous, positively homogeneous, and piecewise linear function of $(s_0, \dots, s_{n-1})$.

By continuity, a bound $L_n^* \geq f$ or identity $L_n^* = f$, with f continuous, need only be proved for $s_{n-1} > 0$. We therefore assume positive sides in the geometric arguments.

2.2. **Chains, clearing, and pins.** Fix a packing of squares Q_i of sides s_i in $[0, L]^2$. For a set of indices I, call $\sum_{i \in I} s_i$ its *side sum*; the same term applies to the corresponding collection $\{Q_i : i \in I\}$. We say that Q_i lies *left of* Q_j when $x_i + s_i \leq x_j$, and *below* Q_j when $y_i + s_i \leq y_j$. Two squares are *x-disjoint* when one lies left of the other, that is, when the interiors of their horizontal projections are disjoint, and *x-overlapping* otherwise. Define y-disjointness and y-overlap using vertical projections. Every pair of squares in a packing is x-disjoint or y-disjoint (possibly both). A *horizontal chain* is a collection of pairwise x-disjoint squares, and a *vertical chain* is a collection of pairwise y-disjoint squares. For positive sides, the left-of and below relations define the *strategy of the packing* [Bad26a, Section 2.2]. Its chains are exactly the nonempty horizontal and vertical chains.

Lemma 2.3 (Chain inequality, [Bad26a, Lemma 3.1])**.** *In a packing of side L, every horizontal or vertical chain has side sum at most L. In particular, $L^*(s) \geq s_0 + s_1$.*

Lemma 2.4 (Column bound, [Bad26a, Proposition 3.7])**.** *For every $n \geq 6$ and $s \in \ell_2^{\downarrow}$,*

$$L^*(s) \geq L_n^*(s) \geq \min\bigl(s_1 + s_3 + s_{n-1},\ P_n(s)\bigr), \quad \text{where } P_n(s) = s_1 + \sum_{i=4}^{n-1} s_i.$$

To prove a lower bound $L^*(s) \geq \Sigma$, suppose a packing has side $L < \Sigma$. A family of indices I, or the corresponding squares, *clears* Σ when $\sum_{i \in I} s_i \geq \Sigma$, and therefore cannot form a horizontal or vertical chain. These exclusions force overlaps, and disjointness of square interiors then gives new chains. For example, suppose Q_a, Q_b are y-disjoint and $\{a, b, e\}$ clears Σ. Then Q_e must y-overlap Q_a or Q_b, since otherwise the three squares would form a vertical chain with side sum exceeding L. Two y-overlapping squares are x-disjoint. We call the forced y-overlap with one of Q_a, Q_b a *vertical pin* of Q_e to the pair Q_a, Q_b. Pins to longer chains and horizontal pins are defined analogously.

We will sometimes translate a square to a wall while keeping the other squares fixed. For instance, a *blocker to the right* of Q_i is a square that lies right of Q_i and y-overlaps it: moving Q_i right would eventually meet it. If there is no such blocker, Q_i can be moved to the right wall without disturbing the packing. Blockers in other directions are defined analogously. Reflections and interchanges of the axes move the entire packing. Because clearing depends only on the side lengths, an exclusion based on a clearing set remains valid during any motion that preserves a packing.

Lemma 2.5 (Weighted-chain comparison)**.** *Fix a packing of squares $Q_0, \dots, Q_{n-1}$ with positive sides, and let $w_0, \dots, w_{n-1}$ be nonnegative weights, not necessarily in decreasing order. If $\sum_{i \in I} w_i \le M$ for every horizontal and vertical chain $\{Q_i : i \in I\}$ of the packing, then squares of side lengths $w_0, \dots, w_{n-1}$ can be packed in $[0, M]^2$. In particular, $\sum_i w_i^2 \le M^2$.*

Proof. Set $w = (w_0, \dots, w_{n-1})$, and let σ be the strategy of the original packing. Then $\ell_\sigma(w) \le M$, and [Bad26a, Lemma 2.4] gives a packing of squares of side lengths $w_0, \dots, w_{n-1}$ in $[0, \ell_\sigma(w)]^2 \subseteq [0, M]^2$. Comparing areas gives the final assertion. □

2.3. **Cornering and covering chains.**

Lemma 2.6 (Cornering criterion, [Bad26a, Lemma 3.5])**.** *Fix a packing of side $L < \Sigma$. Suppose every triple among Q_0, Q_1, Q_2, Q_3 clears Σ, as does every triple $\{0, b, j\}$ with $b \in \{1, 2, 3\}$ and $j \notin \{0, b\}$ in the full packing. Then Q_0 can be translated to a corner while all other squares remain fixed.*

Lemma 2.7 (Relations forced by four large squares, [Bad26a, Lemma 3.6])**.** *Suppose $Q_0 = [0, s_0]^2$ in a packing of $Q_0, \dots, Q_7$ of side $L < \Sigma$. Assume every triple among $0, 1, 2, 3$ clears Σ, as do*

$$\{0, 1, m\}, \qquad \{0, b, m\}, \qquad \{1, b, m\} \quad \text{for } b = 2, 3 \text{ and } m = 4, 5.$$

After possibly interchanging axes, $x_1 \ge s_0$ and $y_1 < s_0$. There are distinct $p, c \in \{2, 3\}$ such that p x-overlaps 0, c x-overlaps 1 and lies above 1, and p lies left of c. Each of $4, 5$ is x-disjoint from both p, c and y-overlaps both. These relations hold in the original full packing, before any translation of Q_1.

Lemma 2.8 (Corner frame, [Bad26a, Lemma 5.2])**.** *Fix a packing of $Q_0, \dots, Q_N$ of side $L < \Sigma$ and an index $m \le N$, and suppose every triple of indices in $\{0, \dots, m\}$ that contains 0 clears Σ. Then Q_0 can be translated to a corner in the packing of $Q_0, \dots, Q_m$ obtained by removing $Q_{m+1}, \dots, Q_N$. After reflection, let $Q_0 = [0, s_0]^2$ and put*

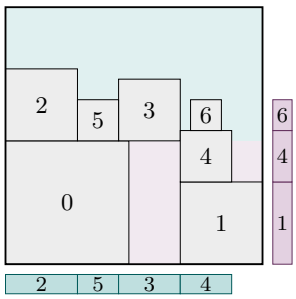
$$\mathcal{R} = \{i \in \{1, \dots, m\} : x_i \ge s_0\}, \qquad \mathcal{U} = \{i \in \{1, \dots, m\} : y_i \ge s_0\}.$$

The arms cover $\{1, \dots, m\}$. The right arm is a vertical chain with pairwise x-overlaps, the upper arm is a horizontal chain with pairwise y-overlaps, and $|\mathcal{R} \cap \mathcal{U}| \le 1$. If Q_0 is already cornered in the packing of $Q_0, \dots, Q_N$, these conclusions require no motion.

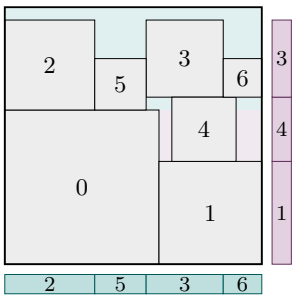
The proof of Proposition 4.1 needs a refinement of [Bad26a, Lemma 5.3] in which the common index of the two covering chains differs from m.

Lemma 2.9 (Perpendicular covering chains)**.** *In the frame of Lemma 2.8, with $m \ge 2$, there are a vertical chain $\mathcal{V}$ and a horizontal chain $\mathcal{H}$ with $\mathcal{V} \cup \mathcal{H} = \{1, \dots, m\}$ and $\mathcal{V} \cap \mathcal{H} = \{j\}$ for some $j \ne m$. In particular, for $m = 6$ some four squares form a chain.*

Proof. If the arms meet at $q \ne m$, they suffice. If either exclusive arm $\mathcal{R} \setminus \mathcal{U}$ or $\mathcal{U} \setminus \mathcal{R}$ is empty, one arm contains every index; pair it with a singleton whose index differs from m.



(a) $r = 4$ right of $u = 3$, $L = 25$
$s = (12, 8, 7, 6, 5, 4, 3)$



(b) $u = 3$ above $r = 4$, $L = 20$
$s = (12, 8, 7, 6, 5, 4, 3)$

FIGURE 2.1. Illustrations of the arm geometry used in the proof of Lemma 2.9, with the arms of Lemma 2.8 sharing Q_6; swapping axes interchanges the two orientations. The side vectors do not satisfy the clearing hypotheses of Lemma 2.8. The bars record the covering chains: $\mathcal{R}$ and $(\mathcal{U} \setminus \{6\}) \cup \{4\}$ in (a), and $(\mathcal{R} \setminus \{6\}) \cup \{3\}$ and $\mathcal{U}$ in (b).

Otherwise let r be topmost in $\mathcal{R} \setminus \mathcal{U}$ and u rightmost in $\mathcal{U} \setminus \mathcal{R}$. Since $x_r \geq s_0 > x_u$ and $y_u \geq s_0 > y_r$, either r lies right of u or u lies above r. Interchanging the axes and the roles of r and u if necessary, assume that r lies right of u; Figure 2.1 illustrates the two orientations. Then $\mathcal{R}$ and $(\mathcal{U} \setminus \mathcal{R}) \cup \{r\}$ are covering chains with common index r. If $r \neq m$, they suffice.

Suppose $r = m$. The arms are disjoint: a common index other than m was already handled, while $m \in \mathcal{R} \setminus \mathcal{U}$. If $\mathcal{R} = \{m\}$, then $\mathcal{U} \cup \{m\}$ is a full horizontal chain; pair it with a singleton whose index differs from m. Otherwise let r' be topmost in $\mathcal{R} \setminus \{m\}$. If r' is right of u, use $\mathcal{R}$ and $\mathcal{U} \cup \{r'\}$. If not, u lies above r'; use $(\mathcal{R} \setminus \{m\}) \cup \{u\}$ and $\mathcal{U} \cup \{m\}$. In either case the common index differs from m. For $m = 6$ the two chain cardinalities sum to seven. Hence one is at least four. □

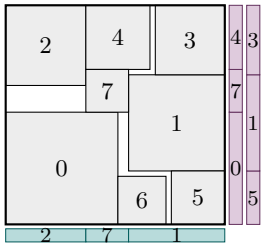
3. Twenty-three strategies

Let $\alpha, \beta, \ldots, \psi$ be the twenty-three maxima of side sums in Table 1.1, regarded as functions of s. Throughout, ω ranges over these twenty-three functions, and $\min_\omega$ denotes the minimum over them. Set $F_8 = \max(L_7^*, \min_\omega \omega)$. The figures and Table 3.1 use the same Greek letters for the corresponding strategies.

Proposition 3.1 (The twenty-three constructions)**.** *For $L \geq \ell_\omega(s) = \max(L_6^*(s), \omega(s))$, the coordinates in the row of ω in Table 3.1 give a packing. Their uniformly valid separation relations define a strategy σ_ω with cost ℓ_ω. Consequently,*

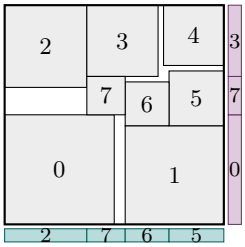
$$L_8^* \leq \min_\omega \ell_\omega = \max\left(L_7^*, \min_\omega \omega\right).$$

Proof. Fix ω and put $g(s) = \max(L_6^*(s), \omega(s))$. Substitution in (2.1) and (2.2) verifies containment and one fixed separation for each pair, uniformly for every $s \in \ell_2^\downarrow$ and $L \geq$



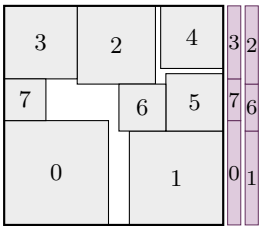
$\alpha, \quad L_8^* = 41$

$s = (21, 18, 15, 13, 12, 10, 9, 8)$



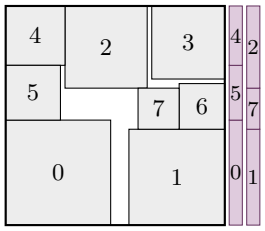
$\beta, \quad L_8^* = 40$

$s = (20, 18, 15, 13, 11, 10, 8, 7)$



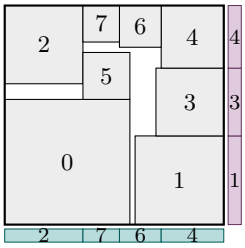
$\gamma, \quad L_8^* = 42$

$s = (20, 18, 15, 14, 12, 11, 9, 8)$



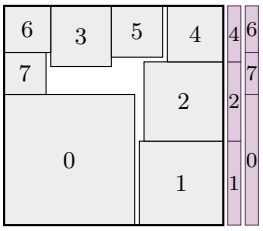
$\delta, \quad L_8^* = 48$

$s = (23, 21, 18, 16, 13, 12, 10, 9)$



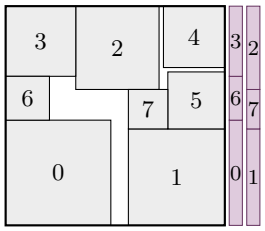
$\varepsilon, \quad L_8^* = 42$

$s = (24, 17, 15, 13, 12, 9, 8, 7)$



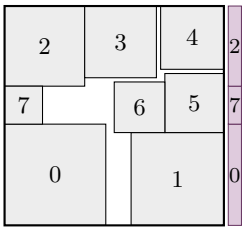
$\zeta, \quad L_8^* = 47$

$s = (28, 18, 17, 13, 12, 11, 10, 9)$



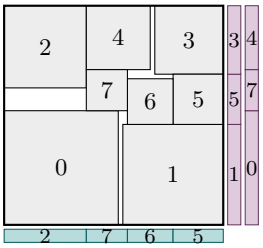
$\eta, \quad L_8^* = 50$

$s = (24, 22, 19, 16, 14, 13, 10, 9)$



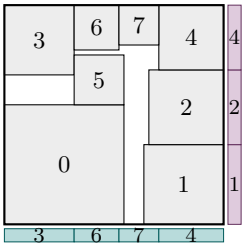
$\theta, \quad L_8^* = 52$

$s = (24, 22, 19, 17, 15, 14, 12, 9)$



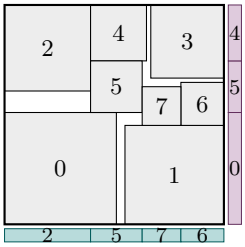
$\iota, \quad L_8^* = 48$

$s = (25, 22, 18, 15, 14, 11, 10, 9)$



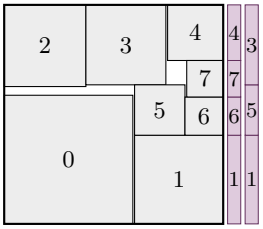
$\kappa, \quad L_8^* = 44$

$s = (24, 16, 15, 14, 13, 10, 9, 8)$



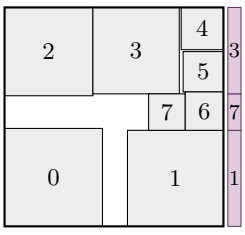
$\lambda, \quad L_8^* = 51$

$s = (26, 23, 20, 17, 13, 12, 10, 9)$



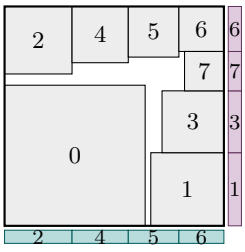
$\mu, \quad L_8^* = 126$

$s = (74, 51, 47, 46, 32, 29, 22, 21)$



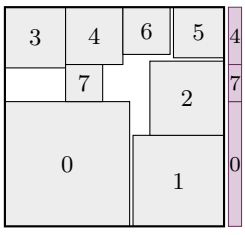
$\nu, \quad L_8^* = 114$

$s = (51, 50, 46, 45, 22, 21, 20, 19)$



$\xi, \quad L_8^* = 39$

$s = (25, 13, 12, 11, 10, 9, 8, 7)$



$o, \quad L_8^* = 65$

$s = (37, 27, 22, 18, 17, 15, 14, 11)$

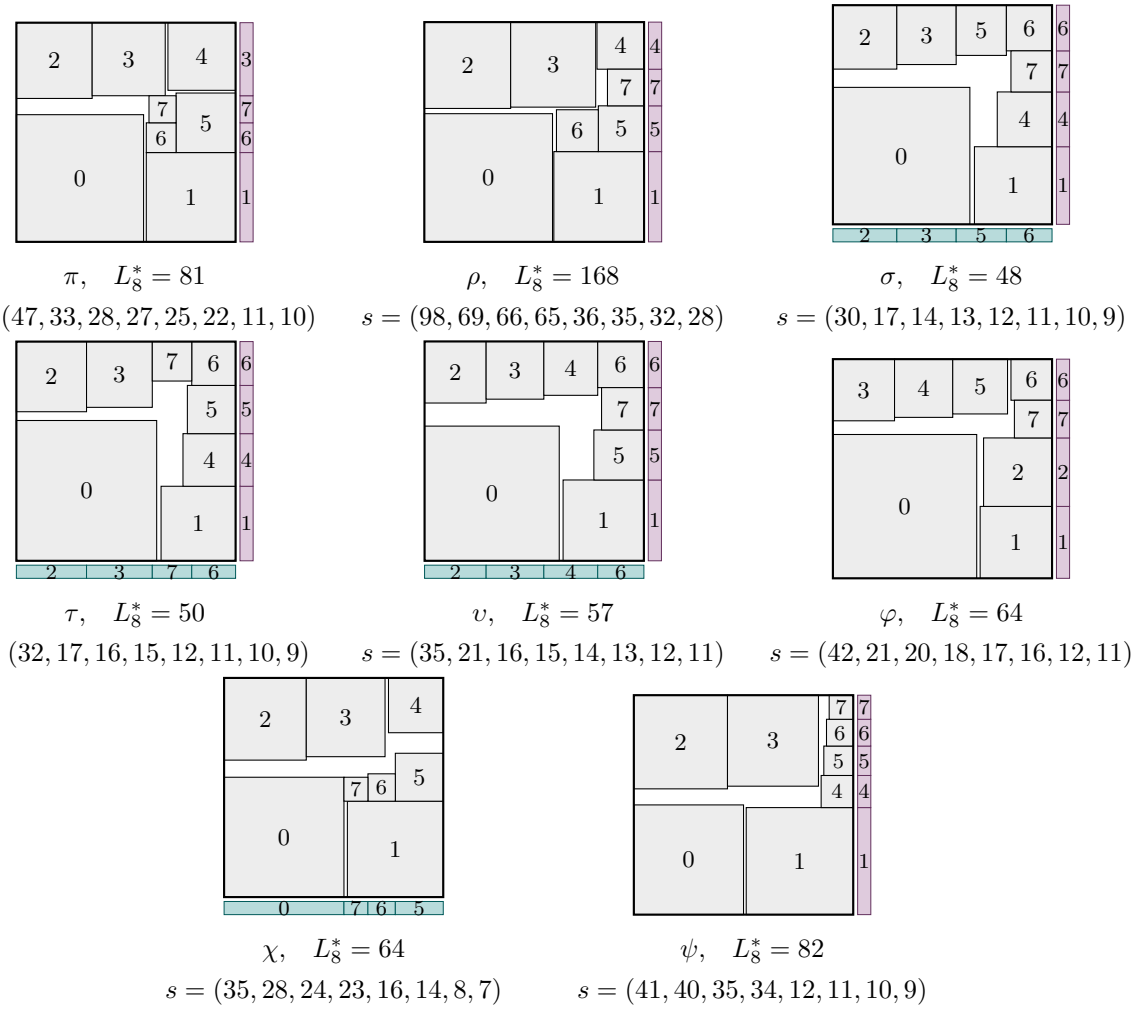


FIGURE 3.1. Placements for the twenty-three strategies, α through o on the previous page and π through ψ here, each at a side vector with $L_8^* = L_7^* + 1$ at which its maximum in Table 1.1 is the only one of the twenty-three equal to L_8^*. Square Q_i carries label i. A teal bar below a panel records the horizontal projections of a chain with side sum L_8^*, and a purple bar at the right records a vertical chain in the same way. The bars show every side sum in the maximum that equals L_8^*. Panels are not to a common scale.

$g(s)$. Each check has the form $aL + \sum_{i=0}^{7} c_i s_i \geq 0$ with $a \geq 0$. Every linear form in $s_0, \ldots, s_7$ satisfies

$$\sum_{i=0}^{7} c_i s_i = \sum_{k=0}^{7} \left(\sum_{i=0}^{k} c_i \right) (s_k - s_{k+1}), \tag{3.1}$$

where $s_8 = 0$ for the truncated tuple. Hence a linear form with nonnegative coefficient prefix sums is nonnegative. For each check, choose a side sum S occurring in g for which the linear form obtained by setting $L = S$ has nonnegative coefficient prefix sums; the

choice is made separately for each containment or separation inequality. Since $L \ge g(s) \ge S$ and $a \ge 0$, this proves the required inequality. For example, in the row of ψ, Q_7 lies above Q_6 because $L - s_7 \ge s_1 + s_4 + s_5 + s_6$, and Q_3 lies left of Q_4 because $s_2 + s_3 \le L - s_4$. These follow from $L \ge \psi(s)$ and $L \ge s_2 + s_3 + s_4$, respectively.

The horizontal and vertical relations consisting of all uniformly valid separations define a strategy σ_ω with $\ell_{\sigma_\omega} \le g$ by [Bad26a, Lemma 3.3]. Each side sum in ω is the side sum of a chain of σ_ω, so $\ell_{\sigma_\omega} \ge \omega$. Lemma 2.2 also gives $\ell_{\sigma_\omega} \ge L_8^* \ge L_6^*$. Hence $\ell_{\sigma_\omega} = g$. The code in [Bad26b] verifies all the containment, separation, and chain checks.

The constructions prove

$$L_8^* \le \min_\omega \max(L_6^*, \omega) = \max\Big(L_6^*, \min_\omega \omega\Big).$$

Deletion gives $L_6^* \le L_7^* \le L_8^*$. Since the common value of the two expressions on the right is at least $L_8^* \ge L_7^*$, replacing L_6^* by L_7^* in the last maximum does not change it. This gives the remaining identities without using the eight-square lower bound. □

Figure 3.1 shows each construction at a side vector with $L_8^* > L_7^*$ at which it alone attains L_8^*. The values of L_8^* there follow from Theorem 1.1.

Deleting Q_7 from a strategy σ for eight squares gives a strategy σ^- for the seven largest squares. The chains of σ^- are the chains of σ that do not contain 7, and removing 7 from a chain of σ leaves a chain of σ^- or the empty set. Hence $\ell_{\sigma^-}(s_0, \dots, s_6) = \ell_\sigma(s_0, \dots, s_6, 0)$, and Proposition 3.1 gives

$$\ell_{\sigma_\omega^-}(s) = \max\big(L_6^*(s),\ \omega(s_0, \dots, s_6, 0)\big). \tag{3.2}$$

Comparing (3.2) with the three seven-square costs in (1.2) shows that $\ell_{\sigma_\omega^-}$ is the cost of center for $\omega = \beta, \theta, \pi, \rho, \chi$, the cost of top for $\omega = o, \varphi$, and the cost of right for $\omega = \nu, \tau, \psi$. For each of the other thirteen strategies, $\ell_{\sigma_\omega^-}$ differs from all three seven-square costs, and at the side vector illustrating σ_ω in Figure 3.1 it equals $L_7^* + 1 = L_8^*$. At the side vector illustrating σ_τ, its restriction also has cost $L_7^* + 1 = L_8^*$, although its cost function is that of right. Thus, fourteen of the twenty-three strategies have restrictions that cost more than L_7^* at the side vectors illustrating them.

TABLE 3.1. Coordinates of the lower left corners of $Q_0, \dots, Q_7$ for the twenty-three strategies, as functions of the tuple s and the side L.

strategy	(x_0, y_0) (x_4, y_4)	(x_1, y_1) (x_5, y_5)	(x_2, y_2) (x_6, y_6)	(x_3, y_3) (x_7, y_7)
α	(0, 0) $(s_2,\ L - s_4)$	$(L - s_1,\ s_5)$ $(L - s_5,\ 0)$	$(0,\ L - s_2)$ $(s_0,\ 0)$	$(L - s_3,\ L - s_3)$ $(s_2,\ s_0)$
β	(0, 0) $(L - s_4,\ L - s_4)$	$(L - s_1,\ 0)$ $(L - s_5,\ s_1)$	$(0,\ L - s_2)$ $(L - s_5 - s_6,\ s_1)$	$(s_2,\ L - s_3)$ $(L - s_5 - s_6 - s_7,\ s_0)$
γ	(0, 0) $(L - s_4,\ L - s_4)$	$(L - s_1,\ 0)$ $(L - s_5,\ s_1)$	$(s_3,\ L - s_2)$ $(L - s_5 - s_6,\ s_1)$	$(0,\ L - s_3)$ $(0,\ s_0)$

Table 3.1 (continued)

strategy	(x_0, y_0) (x_4, y_4)	(x_1, y_1) (x_5, y_5)	(x_2, y_2) (x_6, y_6)	(x_3, y_3) (x_7, y_7)
δ	$(0,\ 0)$ $(0,\ L - s_4)$	$(L - s_1,\ 0)$ $(0,\ s_0)$	$(s_4,\ L - s_2)$ $(L - s_6,\ s_1)$	$(L - s_3,\ L - s_3)$ $(L - s_6 - s_7,\ s_1)$
ε	$(0,\ 0)$ $(L - s_4,\ L - s_4)$	$(L - s_1,\ 0)$ $(s_2,\ s_0)$	$(0,\ L - s_2)$ $(s_2 + s_7,\ L - s_6)$	$(L - s_3,\ s_1)$ $(s_2,\ L - s_7)$
ζ	$(0,\ 0)$ $(L - s_4,\ L - s_4)$	$(L - s_1,\ 0)$ $(s_3 + s_6,\ L - s_5)$	$(L - s_2,\ s_1)$ $(0,\ L - s_6)$	$(s_6,\ L - s_3)$ $(0,\ s_0)$
η	$(0,\ 0)$ $(L - s_4,\ L - s_4)$	$(L - s_1,\ 0)$ $(L - s_5,\ s_1)$	$(s_3,\ L - s_2)$ $(0,\ s_0)$	$(0,\ L - s_3)$ $(L - s_5 - s_7,\ s_1)$
θ	$(0,\ 0)$ $(L - s_4,\ L - s_4)$	$(L - s_1,\ 0)$ $(L - s_5,\ s_1)$	$(0,\ L - s_2)$ $(L - s_5 - s_6,\ s_1)$	$(s_2,\ L - s_3)$ $(0,\ s_0)$
ι	$(0,\ 0)$ $(s_2,\ L - s_4)$	$(L - s_1,\ 0)$ $(L - s_5,\ s_1)$	$(0,\ L - s_2)$ $(L - s_5 - s_6,\ s_1)$	$(L - s_3,\ L - s_3)$ $(s_2,\ s_0)$
κ	$(0,\ 0)$ $(L - s_4,\ L - s_4)$	$(L - s_1,\ 0)$ $(s_3,\ s_0)$	$(L - s_2,\ s_1)$ $(s_3,\ L - s_6)$	$(0,\ L - s_3)$ $(s_3 + s_6,\ L - s_7)$
λ	$(0,\ 0)$ $(s_2,\ L - s_4)$	$(L - s_1,\ 0)$ $(s_2,\ s_0)$	$(0,\ L - s_2)$ $(L - s_6,\ s_1)$	$(L - s_3,\ L - s_3)$ $(L - s_6 - s_7,\ s_1)$
μ	$(0,\ 0)$ $(L - s_4,\ L - s_4)$	$(L - s_1,\ 0)$ $(L - s_5 - s_6,\ s_1)$	$(0,\ L - s_2)$ $(L - s_6,\ s_1)$	$(s_2,\ L - s_3)$ $(L - s_7,\ s_1 + s_6)$
ν	$(0,\ 0)$ $(L - s_4,\ L - s_4)$	$(L - s_1,\ 0)$ $(L - s_5,\ s_1 + s_6)$	$(0,\ L - s_2)$ $(L - s_6,\ s_1)$	$(s_2,\ L - s_3)$ $(L - s_6 - s_7,\ s_1)$
ξ	$(0,\ 0)$ $(s_2,\ L - s_4)$	$(L - s_1,\ 0)$ $(s_2 + s_4,\ L - s_5)$	$(0,\ L - s_2)$ $(L - s_6,\ L - s_6)$	$(L - s_3,\ s_1)$ $(L - s_7,\ s_1 + s_3)$
o	$(0,\ 0)$ $(s_3,\ L - s_4)$	$(L - s_1,\ 0)$ $(L - s_5,\ L - s_5)$	$(L - s_2,\ s_1)$ $(s_3 + s_4,\ L - s_6)$	$(0,\ L - s_3)$ $(s_3,\ L - s_4 - s_7)$
π	$(0,\ 0)$ $(L - s_4,\ L - s_4)$	$(L - s_1,\ 0)$ $(L - s_5,\ s_1)$	$(0,\ L - s_2)$ $(L - s_5 - s_6,\ s_1)$	$(s_2,\ L - s_3)$ $(L - s_5 - s_7,\ s_1 + s_6)$
ρ	$(0,\ 0)$ $(L - s_4,\ L - s_4)$	$(L - s_1,\ 0)$ $(L - s_5,\ s_1)$	$(0,\ L - s_2)$ $(L - s_5 - s_6,\ s_1)$	$(s_2,\ L - s_3)$ $(L - s_7,\ s_1 + s_5)$
σ	$(0,\ 0)$ $(L - s_4,\ s_1)$	$(L - s_1,\ 0)$ $(s_2 + s_3,\ L - s_5)$	$(0,\ L - s_2)$ $(L - s_6,\ L - s_6)$	$(s_2,\ L - s_3)$ $(L - s_7,\ s_1 + s_4)$
τ	$(0,\ 0)$ $(L - s_4,\ s_1)$	$(L - s_1,\ 0)$ $(L - s_5,\ s_1 + s_4)$	$(0,\ L - s_2)$ $(L - s_6,\ L - s_6)$	$(s_2,\ L - s_3)$ $(s_2 + s_3,\ L - s_7)$
υ	$(0,\ 0)$ $(s_2 + s_3,\ L - s_4)$	$(L - s_1,\ 0)$ $(L - s_5,\ s_1)$	$(0,\ L - s_2)$ $(L - s_6,\ L - s_6)$	$(s_2,\ L - s_3)$ $(L - s_7,\ s_1 + s_5)$
φ	$(0,\ 0)$ $(s_3,\ L - s_4)$	$(L - s_1,\ 0)$ $(s_3 + s_4,\ L - s_5)$	$(L - s_2,\ s_1)$ $(L - s_6,\ L - s_6)$	$(0,\ L - s_3)$ $(L - s_7,\ s_1 + s_2)$
χ	$(0,\ 0)$ $(L - s_4,\ L - s_4)$	$(L - s_1,\ 0)$ $(L - s_5,\ s_1)$	$(0,\ L - s_2)$ $(L - s_5 - s_6,\ s_1)$	$(s_2,\ L - s_3)$ $(L - s_5 - s_6 - s_7,\ s_1)$
ψ	$(0,\ 0)$ $(L - s_4,\ s_1)$	$(L - s_1,\ 0)$ $(L - s_5,\ s_1 + s_4)$	$(0,\ L - s_2)$ $(L - s_6,\ s_1 + s_4 + s_5)$	$(s_2,\ L - s_3)$ $(L - s_7,\ L - s_7)$

4. Lower bounds

We prove eight lower bounds $L_8^*(s) \geq b_i(s)$, $i = 1, \ldots, 8$, each given by side sums of $s_0, \ldots, s_7$, in Propositions 4.1–4.11. Section 5.1 combines them with L_7^* and the column bound of Lemma 2.4. Table 4.1 records the notation for recurring side sums.

The bounds $b_2, \ldots, b_8$ are minima of side sums. In their proofs, we set $\Sigma = b_i$ and suppose that a packing has side $L < \Sigma$. A family of squares clears Σ as soon as its side sum is at least one of the side sums in b_i. Since s is nonincreasing, replacing indices in a family by larger ones does not increase its side sum. These proofs follow a common pattern. In the cornering arguments, some of the resulting exclusions allow Q_0 to be moved to a corner and restrict the positions of the next largest squares. Further exclusions and comparisons of side sums then rule out the remaining placements, ending with a chain that clears Σ. The proof of b_3 needs no cornering; it begins with the weighted-chain comparison of Lemma 2.5. The bound b_1 instead combines two perpendicular covering chains with a comparison of partitions.

Table 4.1. Recurring side sums.

$W = s_3 + s_4 + s_5 + s_6$	$G_6 = s_0 + s_3 + s_6$	$J_7 = s_1 + s_2 + s_7$
$H = s_1 + s_2 + s_5$	$G_7 = s_0 + s_3 + s_7$	$J_{67} = s_1 + s_2 + s_6 + s_7$
$C_0 = s_0 + s_5 + s_6$	$K = s_0 + s_2 + s_7$	$M = s_1 + s_4 + s_6 + s_7$
$C_3 = s_1 + s_3 + s_6$	$T = s_0 + s_6 + s_7$	$P_\pi = s_1 + s_3 + s_6 + s_7$
$R = s_1 + s_4 + s_5 + s_6$	$U = s_0 + s_4 + s_5$	$P_\rho = s_1 + s_4 + s_5 + s_7$
$D = s_1 + s_3 + s_7$	$H_4 = s_1 + s_3 + s_4$	$P_\chi = s_0 + s_5 + s_6 + s_7$
$E = s_0 + s_4 + s_7$	$H_5 = s_1 + s_3 + s_5$	$V = s_2 + s_5 + s_6 + s_7$
$P = s_1 + s_4 + s_5 + s_6 + s_7$	$J_4 = s_1 + s_2 + s_4$	$X = s_3 + s_4 + s_6 + s_7$
	$J_6 = s_1 + s_2 + s_6$	$Y = s_2 + s_4 + s_6 + s_7$

4.1. **The balanced-partition bound.**

Proposition 4.1. *For every $s \in \ell_2^\downarrow$,*

$$L_8^*(s) \geq b_1(s) := \min\bigl(T,\ \mathcal{K}(s)\bigr),$$

where $T = s_0 + s_6 + s_7$ and

$$\mathcal{K}(s) = \min_{\substack{I \sqcup J = \{1,2,3,4,5,7\} \\ |I| = |J| = 3}} \max\left(s_6 + \sum_{i \in I} s_i,\ s_6 + \sum_{j \in J} s_j \right).$$

Proof. Suppose $L < \Sigma := b_1$. A triple containing Q_0 has side sum at least T, because its other two squares are no smaller than Q_6 and Q_7. Thus, every such triple clears Σ. Apply Lemmas 2.8 and 2.9 with $m = 7$. They give perpendicular covering chains with common index $j \leq 6$.

If $j \neq 6$, interchange the labels j and 6 in the two index sets. The family containing both has unchanged side sum, while the other sum decreases by $s_j - s_6 \geq 0$. This is a statement about sums only; the relabelled families need not remain geometric chains. Removing the new common index 6 gives a partition

$$I \sqcup J = \{1, 2, 3, 4, 5, 7\}, \qquad s_6 + \sum_{i \in I} s_i \leq L, \quad s_6 + \sum_{i \in J} s_i \leq L.$$

Because $s_0 + s_1 \leq L < T$ by Lemma 2.3, we have $s_1 < s_6 + s_7 \leq s_5 + s_7$. Among the six numbers indexed by $1, 2, 3, 4, 5, 7$, every pair therefore sums to more than every singleton.

Under this pair–singleton property, any two-part partition can be replaced by a $3 + 3$ partition without increasing its maximum part-sum. For a $2 + 4$ partition, write the first two entries as a, b and split the other four into pairs of sums u, v. Then $a+u, b+v \leq u+v$. For a $1 + 5$ partition, join the singleton to two entries of the other part. Since the three remaining entries sum to at least the singleton, neither new triple exceeds the old five-term sum. The $0 + 6$ case is immediate. The resulting $3 + 3$ partition gives $\mathcal{K} \leq L$, contrary to $L < \Sigma$. □

4.2. **The bridge bound.**

Proposition 4.2. *For every $s \in \ell_2^{\downarrow}$,*

$$L_8^*(s) \geq b_2(s) := \min(C_0,\ E,\ J_4,\ J_{67}),$$

where

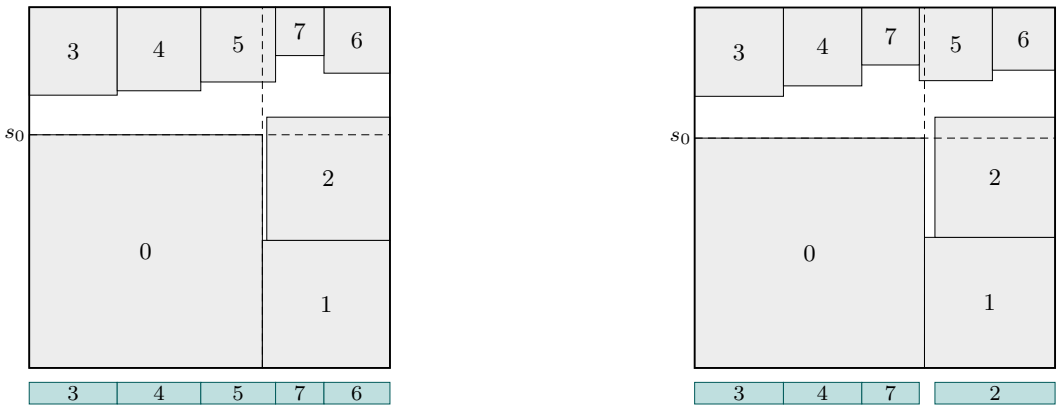
$$C_0 = s_0 + s_5 + s_6, \quad E = s_0 + s_4 + s_7, \quad J_4 = s_1 + s_2 + s_4, \quad J_{67} = s_1 + s_2 + s_6 + s_7.$$

Proof. Fix $\Sigma = b_2$ and suppose $L < \Sigma$. Since $L \geq s_0 + s_1$, the inequalities $L < C_0$ and $L < E$ give

$$s_5 + s_6 > s_1, \qquad s_4 + s_7 > s_1. \tag{4.1}$$

Any three of $Q_0, \ldots, Q_3$ have side sum at least $s_1 + s_2 + s_3 \geq J_4$. A triple consisting of Q_0, one of Q_1, Q_2, Q_3, and any third square has side sum at least $s_0 + s_3 + s_7 \geq E$. All these triples clear Σ, and Lemma 2.6 therefore corners Q_0 with all eight squares present.

We first show that $Q_1, \ldots, Q_6$ contain a four-square chain, and that it must be $\{Q_3, Q_4, Q_5, Q_6\}$. Among $Q_0, \ldots, Q_6$, every triple containing Q_0 has side sum at least C_0 and clears Σ. Since Q_0 is already cornered, the no-motion clause of Lemma 2.8 applies to the prefix $Q_0, \ldots, Q_6$. Every four-subset of $\{1, \ldots, 6\}$ except $\mathcal{W} = \{3, 4, 5, 6\}$ has sum at least $s_2 + s_4 + s_5 + s_6 > J_4 \geq \Sigma$ by (4.1). Lemma 2.9 forces a four-chain, which must therefore be $\mathcal{W}$. Interchange axes so that it is horizontal. Members of the right arm pairwise x-overlap, whereas members of $\mathcal{W}$ are pairwise x-disjoint. Hence at most one member of $\mathcal{W}$ is in the right arm, and at least three are in the upper arm. If 1 or 2 were in the upper arm, it and those three would give a clearing four-chain different from $\mathcal{W}$. Thus, $1, 2$ lie in the exclusive right arm, form a vertical chain, and have lower edges below s_0. Let t be their upper square.



(a) Q_7 right of Q_0, $L = 82$
$s = (53, 29, 28, 20, 19, 17, 15, 11)$

(b) Q_7 above Q_0, $L = 69$
$s = (44, 25, 23, 17, 15, 14, 12, 11)$

FIGURE 4.1. The two placements of Q_7 in the proof of Proposition 4.2; the dashed lines are $x = s_0$ and $y = s_0$. Here L is less than C_0, E, and J_{67}, which are 85, 83, 83 in (a) and 70, 70, 71 in (b), but $J_4 \le L$. Squares Q_1, Q_2 form the exclusive right arm with upper square $t = 2$, and Q_3, Q_4 lie left of Q_t. Square Q_6 lies in the right arm, and the bridge $u = 5$ satisfies (4.2). (a) Q_7 lies right of Q_0 and above Q_t; the bar records the horizontal chain $\{3, 4, 5, 6, 7\}$. (b) Q_7 lies above Q_0 and is x-disjoint from Q_t; the bar records the horizontal chain $3, 4, 7, t$. If the remaining inequality $L < J_4$ also held, each indicated chain would have side sum greater than L by (4.3), a contradiction.

The triples $1, 2, 3$ and $1, 2, 4$ have side sum at least J_4. They clear Σ and force $3, 4$ into the exclusive upper arm. Neither can x-overlap t: it would lie above t and give the corresponding clearing vertical triple with $1, 2$. Since $x_3, x_4 < s_0 \le x_t$, both lie left of t.

Each of $5, 6$ must x-overlap t, or else it and $t, 3, 4$ would be a clearing horizontal quadruple. The two squares $5, 6$ are x-disjoint from one another. Exactly one belongs to the right arm. There is at most one by the arm-overlap property. If neither did, both would start left of s_0 and end right of $x_t \ge s_0$, contradicting their x-disjointness. Denote the other, exclusive upper-arm square by u. Then

$$x_u < s_0 \le x_t < x_u + s_u, \tag{4.2}$$

and u lies above t. We call Q_u the bridge: it crosses the line $x = s_0$ and reaches past the left edge of Q_t. The right-arm member of $\{5, 6\}$ also lies above t. Otherwise it would lie below t, and since u lies above t, the four squares $1, 2, 5, 6$ would be a vertical chain of sum at least J_{67}. Both $5, 6$ now lie above t, and the same sum shows that they y-overlap.

It remains to place the eighth square. Figure 4.1 shows the two placements of Q_7 treated below. Two further side sums exceed $J_4 \ge \Sigma$. With $W = s_3 + s_4 + s_5 + s_6$,

$$s_2 + s_3 + s_4 + s_7 > J_4, \qquad W + s_7 > J_4. \tag{4.3}$$

For the first, $s_3 + s_7 \geq s_4 + s_7 > s_1$. For the second, subtract J_4; the difference is $(s_5 + s_6 - s_1) + (s_3 + s_7 - s_2) > 0$ by (4.1), because s is nonincreasing.

If 7 is x-disjoint from 0, it lies right of 0. The horizontal pins $0, 1, 7$ and $0, 2, 7$ make it x-overlap both $1, 2$. Hence it is y-disjoint from both. For $k = 5, 6$, the quadruple $1, 2, k, 7$ has side sum at least J_{67}, giving a vertical pin of 7 to the chain $1, 2, k$. Thus, 7 y-overlaps both $5, 6$, and is x-disjoint from both. In particular it lies right of u, since $x_7 \geq s_0 > x_u$. By (4.2), $x_7 > x_t$. Hence it is also x-disjoint from $3, 4$, which lie left of t. The five squares $3, 4, 5, 6, 7$ now form a horizontal chain that clears Σ by (4.3).

Otherwise 7 x-overlaps 0 and lies above it. The triples $0, 3, 7$ and $0, 4, 7$ have side sum at least E, and their vertical pins make 7 x-disjoint from $3, 4$. If it is also x-disjoint from t, the horizontal chain $3, 4, 7, t$ has side sum at least $s_2 + s_3 + s_4 + s_7$, which exceeds J_4 by (4.3). If not, it is above t, since $y_7 \geq s_0 > y_t$, and hence y-disjoint from $1, 2$. The same pins, from the quadruples $1, 2, k, 7$, make it x-disjoint from $5, 6$. Again $3, 4, 5, 6, 7$ form the forbidden horizontal five-chain. This proves the bound. □

4.3. **The weighted-chain bound.**

Proposition 4.3. *For every $s \in \ell_2^{\downarrow}$,*

$$L_8^*(s) \geq b_3(s) := \min(J_7,\ V),$$

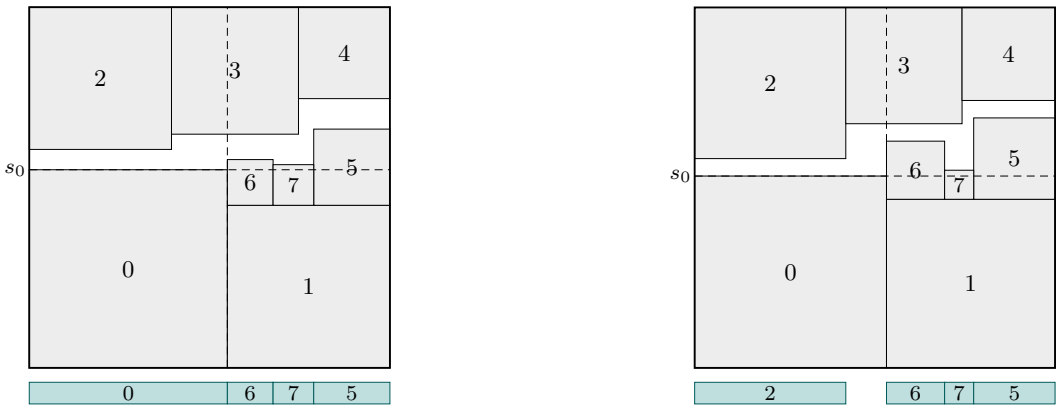
where $J_7 = s_1 + s_2 + s_7$ and $V = s_2 + s_5 + s_6 + s_7$.

Proof. Suppose $L < \Sigma := b_3$. For this proof call $0, 1, 2$ large and $3, 4, 5, 6, 7$ small. A chain containing two large squares and a third square, or one large square and three small squares, has side sum at least J_7 or V, respectively, and clears Σ.

Give the large squares weight two and the small squares weight one. If every chain had weight at most four, Lemma 2.5 would give $3 \cdot 2^2 + 5 \cdot 1^2 = 17 \leq 4^2 = 16$, a contradiction. Thus, some chain has weight at least five. If it contains a large square, it contains one of the two forbidden types above. Hence all five small squares form a chain. Interchange axes so that it is horizontal, and let q be its middle square in left-to-right order.

Every large square x-overlaps q. Otherwise it lies on one side of q and is x-disjoint from at least three small squares, giving a clearing horizontal chain. Thus, q is y-disjoint from each large square. Two large squares cannot lie on opposite sides of q vertically, since they would form a clearing triple with it. Nor can two on the same side be y-disjoint from one another, for the same reason. The three large squares therefore y-overlap pairwise and form a clearing horizontal triple. This contradiction proves the bound. □

4.4. **Two bounds from a right-hand stack.** In this subsection $Q_0, \ldots, Q_3$ are large, Q_4, Q_5 are medium, and Q_6, Q_7 are small. We first show that the medium squares Q_4, Q_5 form a vertical stack on the right, above Q_1; this is the common part of the two proofs. Figure 4.2 shows the configurations that it produces.



(a) Proposition 4.5, $L = 71$
$s = (39, 32, 28, 25, 18, 15, 9, 8)$

(b) Proposition 4.7, $L = 62$
$s = (33, 29, 26, 20, 16, 14, 10, 5)$

FIGURE 4.2. The right-hand stack of Lemma 4.4, with $p = 2$, $c = 3$, and medium squares $b = 5$ (lower) and $t = 4$ (upper); the dashed lines are $x = s_0$ and $y = s_0$. In both panels every family in the hypotheses of the lemma has side sum greater than L, and Q_6, Q_7 lie above Q_1 and below Q_3, each y-overlapping Q_b. (a) $L = 71$ is less than $G_7 = 72$, $U = 72$, $H_5 = 72$, $P_\pi = 74$, and $P_\rho = 73$, but $P_\chi = 71$; the bar records the chain $0, 6, 7, b$ that ends the proof of Proposition 4.5. (b) $L = 62$ is less than $K = 64$, $G_6 = 63$, $U = 63$, $H_5 = 63$, and $J_6 = 65$, but $V = 55$; the bar records the chain $\{2, b, 6, 7\}$ that ends the proof of Proposition 4.7.

Lemma 4.4 (Right-hand stack)**.** *Suppose Q_0 is cornered, the hypotheses of Lemma 2.7 hold, and the following additional families clear Σ:*

$$\{0, 1, e\}\ (e = 6, 7), \qquad \{2, 3, 4, 5\}, \qquad \{0, 4, 5\}, \qquad \{1, 4, 5, e\}\ (e = 6, 7).$$

Use the labels $p, c \in \{2, 3\}$ from that lemma. Then Q_4, Q_5 form a vertical stack with lower square Q_b and upper square Q_t, both lie right of Q_c and above the original Q_1, and

$$x_p + s_p \le x_c < s_0 \le y_c. \tag{4.4}$$

Square Q_1 can be lowered to the floor in the full eight-square packing. In the resulting packing every square other than Q_0, Q_1 lies above Q_1.

Proof. Lemma 2.7 gives the relations among the fixed squares $p, c, 4, 5$. If $4, 5$ were x-disjoint, they and p, c would form a clearing horizontal quadruple. Thus, $4, 5$ x-overlap and form a vertical stack with lower square b and upper square t.

If $y_b \ge s_0$, the vertical chain $0, b, t$ clears Σ. Hence $y_b < s_0$ and $x_b \ge s_0$. By Lemma 2.7, b and c are x-disjoint. If c were right of b, the chain $0, b, c$ would clear Σ. Thus, b lies right of c. Since t x-overlaps b and is x-disjoint from c, it too lies right of c. If $x_c \ge s_0$, then the chain $0, c, b$ would clear Σ. Hence $x_c < s_0$. Since $0 \le x_c < s_0$, square c x-overlaps 0, and hence $y_c \ge s_0$. Together with p left of c, this proves (4.4).

For $m = 4, 5$, we have $x_m \geq x_c + s_c > x_1$. If m were x-disjoint from 1, it would lie right of 1 and form a clearing horizontal chain $0, 1, m$. Therefore, m x-overlaps 1 and is y-disjoint from it. It cannot lie below 1, because $m, 1, c$ would be a vertical chain clearing Σ. Both medium squares thus lie above the original 1.

Squares 0 and p are x-disjoint from 1, the second by (4.4) and $x_1 \geq s_0$, while $c, 4, 5$ lie above 1. A downward blocker of 1 can therefore only be 6 or 7. Such a blocker e would complete the clearing vertical chain $e, 1, 4, 5$. Thus, no blocker exists and 1 can be lowered in the full packing. For every other index $j \neq 0, 1$, the horizontal pin to the pair $0, 1$, with 1 now on the floor, forces x-overlap with at least one of them. It follows that $y_j \geq s_1$, using $s_0 \geq s_1$. □

The first application of Lemma 4.4 is a bound by six side sums.

Proposition 4.5. *For every $s \in \ell_2^{\downarrow}$,*

$$L_8^*(s) \geq b_4(s) := \min(G_7,\ U,\ H_5,\ P_\pi,\ P_\rho,\ P_\chi),$$

where

$$G_7 = s_0 + s_3 + s_7, \quad U = s_0 + s_4 + s_5, \quad H_5 = s_1 + s_3 + s_5,$$
$$P_\pi = s_1 + s_3 + s_6 + s_7, \quad P_\rho = s_1 + s_4 + s_5 + s_7, \quad P_\chi = s_0 + s_5 + s_6 + s_7.$$

Proof. Set $\Sigma = b_4$ and suppose $L < \Sigma$. Since $G_7 > L \geq s_0 + s_1$, we have $s_3 + s_7 > s_1$. Hence $Z := s_2 + s_3 + s_5 + s_7 = H_5 + (s_2 + s_7 - s_1) > H_5 \geq \Sigma$. Thus, $\{2, 3, 4, 5\}$ and every $\{2, 3, m, e\}$ with $m = 4, 5$ and $e = 6, 7$ clear Σ. Every triple of the four large squares has side sum at least H_5, and every triple $\{0, b, j\}$ with $b = 1, 2, 3$ has side sum at least G_7. These triples clear Σ as well, and Lemma 2.6 corners 0 in the full packing. The hypotheses of Lemma 2.7 follow from G_7, H_5. The other hypotheses of Lemma 4.4 are supplied by Z, U, P_ρ. Apply that lemma and lower Q_1 to the floor, keeping the other squares fixed.

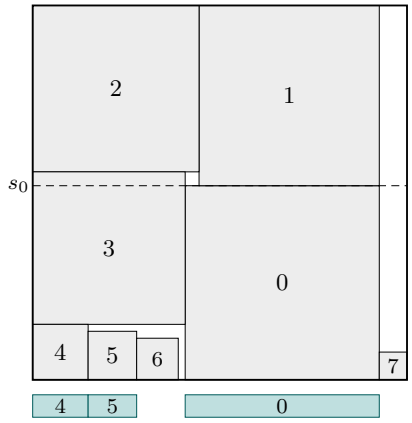
We next show that $y_6, y_7 < s_0$. Both p and c lie above 0: p x-overlaps 0 by Lemma 2.7, and $y_c \geq s_0$ by (4.4). For $e = 6, 7$ we already know $y_e \geq s_1$. If $y_e \geq s_0$, the vertical pins to $0, p$ and $0, c$ force e to y-overlap both p, c, and hence be x-disjoint from them. If it were also x-disjoint from either medium square m, the quadruple p, c, m, e would be a horizontal chain that clears Σ. Thus, it x-overlaps both $4, 5$ and is y-disjoint from both. All three lie above 1, and the vertical chain $1, 4, 5, e$ has side sum at least P_ρ. Hence $y_e < s_0$ and $x_e \geq s_0$.

By (4.4), e is x-disjoint from p. The vertical pin to the chain $1, 4, 5$ makes e y-overlap a medium square m, hence be x-disjoint from it. If it were x-disjoint from c as well, the quadruple p, c, m, e would clear Σ. Thus, e x-overlaps c and is y-disjoint from it. Since $y_e < s_0 \leq y_c$, it lies below c.

Finally, we find a chain that clears Σ. Both $6, 7$ lie above 1 and below c. If they were y-disjoint, the vertical chain $1, 6, 7, c$ would have sum at least P_π. Thus, they y-overlap and are x-disjoint. Let b, t again be the lower and upper medium squares. Since square b y-overlaps c, we have $y_t \geq y_b + s_b > y_c$. Neither small square, whose top edge is at most y_c, can y-overlap t. Its pin to $1, 4, 5$ must therefore be to b, making it x-disjoint from b.

Moreover it lies left of b, since it x-overlaps c and b lies right of c. Finally $0, 6, 7, b$ form a horizontal chain with side sum at least $P_\chi \geq \Sigma$, a contradiction. □

The second application of Lemma 4.4 requires a different cornering argument. Since the next proof does not assume $G_7 \geq \Sigma$, the pair Q_3, Q_7 requires separate treatment. Figure 4.3 shows the configuration excluded in the proof of the next lemma.



Lemma 4.6, $L = 54$
$s = (28, 26, 24, 22, 8, 7, 6, 4)$

FIGURE 4.3. The double blockage excluded in the proof of Lemma 4.6; the dashed line is $y = s_0$. Here $L = 54$ is less than $K = 56$, $G_6 = 56$, $H_5 = 55$, $J_6 = 56$, and $s_1 + s_2 + s_3 = 72$, but not less than $U = 43$ or $V = 41$. Square Q_0 is on the floor, blocked by Q_3 on the left and Q_7 on the right. Squares Q_1, Q_2 lie above Q_0, with $a = 2$ and $b = 1$ in the notation of the proof, and Q_3 lies below Q_2. Squares Q_4, Q_5 lie left of Q_0 and below Q_3, and x-overlap Q_3. The bar records the horizontal chain $4, 5, 0$, excluded by U.

Lemma 4.6 (Exceptional pincer)**.** *Suppose* $L < \Sigma \leq \min(K, G_6, U, H_5, J_6, V)$*, where* $K = s_0 + s_2 + s_7$*,* $G_6 = s_0 + s_3 + s_6$*, and* $J_6 = s_1 + s_2 + s_6$*. Then* Q_0 *can be translated to a corner in the full packing.*

Proof. Because s is nonincreasing, the following triples clear Σ:

$\{0, 1, e\},\ \{0, 2, e\}$	for every distinct third index e,
$\{0, 3, e\}$	for $e \leq 6,\ e \neq 0, 3$,
$\{1, 2, e\}$	for $e \leq 6,\ e \neq 1, 2$,
$\{1, 3, m\}$	for $m = 4, 5$.

Their side sums are at least K, G_6, J_6, and H_5, respectively. Every triple of the four large squares clears Σ as well, and so does the quadruple $2, 3, 4, 5$, whose side sum is at least V.

Choose an axis on which $0, 1$ are disjoint and move 0 away from 1 to the corresponding wall. A blocker would form a clearing triple with $0, 1$. Interchange axes and reflect so

that $y_0 = 0$. If Q_0 has no blocker on its left or no blocker on its right, move it horizontally to the corresponding corner. Otherwise there is a blocker on each side.

Neither 1 nor 2 can be x-disjoint from 0: it would form a clearing horizontal triple with 0 and the opposite blocker. Thus, $1, 2$ x-overlap 0 and lie above it. They cannot be y-disjoint, because $0, 1, 2$ would clear Σ; they are therefore x-disjoint. Square 3 must be x-disjoint from 0. Otherwise it lies above 0, and its vertical pins to $0, 1$ and $0, 2$ make it x-disjoint from both $1, 2$, giving the clearing horizontal triple $1, 2, 3$. Reflect horizontally so that 3 lies left of 0, and let a, b be the left-to-right order of $1, 2$. A blocker with index $j \leq 6$ on the opposite side would give the horizontal chain $3, 0, j$, which clears Σ. Hence the blocker on that side is 7.

Square 3 x-overlaps a, or else $3, a, b$ form a clearing horizontal triple. Thus, it is y-disjoint from a. It cannot lie above a, or be y-disjoint above 0, because $0, a, 3$ would then be a vertical chain. Hence 3 lies below a and y-overlaps 0.

For $m = 4, 5$, square m must also y-overlap 0. Otherwise it lies above 0, and pins to $0, a$ and $0, b$ make it x-disjoint from both, giving the clearing horizontal triple $1, 2, m$. Thus, m is x-disjoint from 0. Its horizontal pin to $0, 3$ forces it to x-overlap 3. Hence it lies left of 0, and it lies left of b because $x_b \geq x_a + s_a > x_0$. The forbidden triple a, b, m then forces a horizontal overlap with a. Therefore, m is y-disjoint from both $3, a$, and lies below a because $y_m < s_0 \leq y_a$.

If $a = 1$, the vertical triple $1, 3, m$ clears Σ. Hence $a = 2$. Now $2, 3$ are y-disjoint from each other and from both $4, 5$. If $4, 5$ were y-disjoint, the quadruple $2, 3, 4, 5$ would be a vertical chain that clears Σ. Otherwise they y-overlap and are x-disjoint. Both lie left of 0, and the horizontal triple $0, 4, 5$, with side sum U, clears Σ. This excludes the double blockage and completes the cornering argument. □

Proposition 4.7. *For every* $s \in \ell_2^{\downarrow}$,

$$L_8^*(s) \geq b_5(s) := \min(K,\ G_6,\ U,\ H_5,\ J_6,\ V).$$

Proof. Set $\Sigma = b_5$ and suppose $L < \Sigma$. The triple exclusions in Lemma 4.6 hold. In addition, each of the following families has side sum at least V:

$$(4.5)\ \ \{2, 3, 4, 5\}, \quad \{2, 3, m, e\}\ (m = 4, 5;\ e = 6, 7), \quad \{1, 4, 5, e\}\ (e = 6, 7), \quad \{1, 3, 6, 7\}.$$

For example, the least sums in the last two families are $s_1 + s_4 + s_5 + s_7 \geq V$ and $s_1 + s_3 + s_6 + s_7 \geq V$.

Lemma 4.6 corners 0. The hypotheses of Lemma 2.7 follow from the displayed triple exclusions, and those of Lemma 4.4 follow from U and (4.5). Thus, we obtain the right-hand stack, (4.4), and Q_1 lowered to the floor in the full packing.

We next use square 6 to identify the labels p, c of the large squares. All squares other than $0, 1$ lie above 1. The triple $1, 2, 6$ clears Σ, and its vertical pin makes 6 y-overlap 2, hence be x-disjoint from it. The pin to $1, 4, 5$ makes it x-disjoint from a medium square m. If it were also x-disjoint from 3, the quadruple $2, 3, m, 6$ would clear Σ. Thus, 6 x-overlaps 3 and is y-disjoint from it.

If $y_6 \geq s_0$, the vertical pins to $0, 2$ and $0, 3$, from the clearing triples $0, 2, 6$ and $0, 3, 6$, make it x-disjoint from both, a contradiction. Hence $y_6 < s_0$ and $x_6 \geq s_0$. By (4.4), 6 is x-disjoint from the left large square p. Since it x-overlaps 3, we must have $p = 2, c = 3$. Moreover 6 lies below 3.

Finally, we locate square 7 and find a chain that clears Σ. Square 7 is x-disjoint from 2. If $y_7 < s_0$, it lies right of 0, while $2 = p$ lies entirely left of $x = s_0$ by (4.4). Otherwise the vertical pin to $0, 2$, from the clearing triple $0, 2, 7$, makes it y-overlap 2. The pin to $1, 4, 5$ makes 7 x-disjoint from a medium square m. If it were also x-disjoint from 3, the quadruple $2, 3, m, 7$ would clear Σ. Thus, both $6, 7$ x-overlap 3 and are y-disjoint from it. They cannot be y-disjoint from each other, since $1, 3, 6, 7$ clears Σ. They therefore y-overlap, are x-disjoint, and lie on the same vertical side of 3, hence below it, as 6 does.

Since the lower medium square b y-overlaps 3, the upper medium square t satisfies $y_t \geq y_b + s_b > y_3$. Neither small square can y-overlap t. The pins to $1, 4, 5$ must therefore be to b, making both small squares x-disjoint from it. The horizontal chain $\{2, b, 6, 7\}$ has sum at least $V \geq \Sigma$, the final contradiction. □

4.5. **Three bounds from a four-square core.** Call the elements of $\mathcal{C} = \{1, 2, 3, 4\}$ the core indices and those of $\mathcal{S} = \{5, 6, 7\}$ the small indices. These names are local to this subsection; the largest square Q_0 is kept separate.

We use two families of chain exclusions. In all applications below they follow from side-sum inequalities, and hence hold in every packing of the given side lengths in $[0, L]^2$:

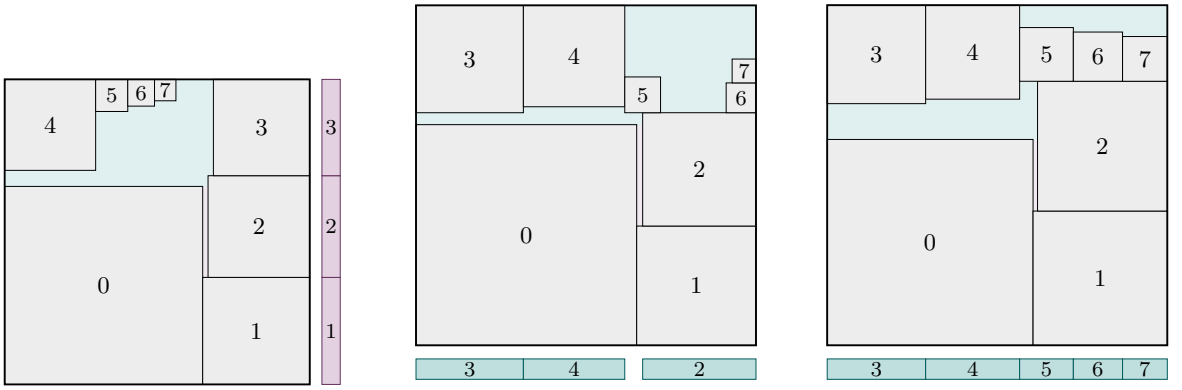
(C): No chain contains Q_0, Q_b, Q_j, where $b \in \mathcal{C}$ and $j \notin \{0, b\}$.
(Q): No four-square chain contains at least three indices from $\mathcal{C}$.

Figure 4.4 illustrates the two alternatives in the next lemma.

Lemma 4.8 (Cornering and the two frame alternatives)**.** *Suppose (C) and (Q) hold for every packing of the given side lengths in $[0, L]^2$. Then Q_0 can be translated to a corner while all other squares remain fixed. After reflections set $Q_0 = [0, s_0]^2$, $\mathcal{R} = \{i \geq 1 : x_i \geq s_0\}$, and $\mathcal{U} = \{i \geq 1 : y_i \geq s_0\}$. Up to interchanging axes, one of the following holds.*

I. *Three core squares form the right-arm vertical chain. Let t be its topmost index and k the remaining core index. All three small squares lie in $\mathcal{U} \setminus \mathcal{R}$. Squares Q_k, Q_t are x-disjoint from each other and from all three small squares.*
II. *Write the right-arm core squares as b, t, bottom to top, and the upper-arm core squares as p, q, left to right. The axes may be chosen so that Q_t lies right of Q_q. Then p, q, t is a horizontal three-core chain. Every small square x-overlaps Q_t and is y-disjoint from both Q_b, Q_t.*

If, in Alternative II, no chain can contain b, t and two small indices, then the three small squares y-overlap pairwise and form a horizontal chain. Its middle and rightmost squares are x-disjoint from both Q_p, Q_q. Moreover, either all three small squares belong to $\mathcal{R}$, or all three lie above Q_t.



(a) Alternative I, $L = 57$
$s = (37, 20, 19, 18, 17, 6, 5, 4)$

(b) Alternative II, $L = 57$
$s = (37, 20, 19, 18, 17, 6, 5, 4)$

(c) Alternative II, $L = 76$
$s = (46, 30, 29, 22, 21, 12, 11, 10)$

FIGURE 4.4. The frame alternatives of Lemma 4.8; the shaded regions are $x \geq s_0$ and $y \geq s_0$. In (a) and (b), $L = 57$ is less than $s_0 + s_4 + s_7 = 58$ and $s_2 + s_3 + s_4 + s_7 = 58$, so (C) and (Q) hold. (a) Alternative I with $t = 3$ and $k = 4$; the bar records the right-arm chain $1, 2, 3$. (b) Alternative II with $b = 1$, $t = 2$, $p = 3$, and $q = 4$; the bar records the horizontal chain p, q, t. (c) Alternative II with $L = 76$ less than $s_0 + s_4 + s_7 = 77$, $s_2 + s_3 + s_4 + s_7 = 82$, and $s_1 + s_2 + s_6 + s_7 = 80$, so no chain contains b, t and two small squares. The small squares form a horizontal chain above Q_t, and Q_6, Q_7 lie right of Q_p, Q_q. The bar records the chain $3, 4, 5, 6, 7$ that ends the proofs of Propositions 4.10 and 4.11.

Proof. Choose an axis on which Q_0, Q_1 are disjoint. Move Q_0 away from Q_1 toward the corresponding wall. A blocker would make a chain with Q_0, Q_1, contrary to (C). Interchange axes and reflect so that the wall reached is the bottom wall, $y_0 = 0$.

If Q_0 has no blocker on its left or no blocker on its right, move it horizontally to the corresponding corner. Otherwise there is a blocker on each side. Any core square x-disjoint from Q_0 would form a chain with Q_0 and a blocker on the opposite side. By (C), every core square therefore x-overlaps Q_0, and hence lies above it. Any two core squares that were y-disjoint would form a chain with Q_0, again contrary to (C). The four core squares consequently y-overlap pairwise and form a horizontal chain, contradicting (Q).

After cornering, the sets $\mathcal{R}$ and $\mathcal{U}$ cover $\{1, \ldots, 7\}$. A core square in $\mathcal{R}$ x-overlaps every other square in $\mathcal{R}$: otherwise those two squares and Q_0 violate (C). Thus, it is y-disjoint from all of them. Symmetrically, a core square in $\mathcal{U}$ y-overlaps every other square in $\mathcal{U}$, and is x-disjoint from it. Unlike in Lemma 2.8, the arms need not be chains: (C) gives no exclusion for a triple consisting of Q_0 and two small squares.

Let $r = |\mathcal{C} \cap \mathcal{R}|$ and $u = |\mathcal{C} \cap \mathcal{U}|$. The core squares in either arm form a chain. By (Q), $r, u \leq 3$. Also at most one core square belongs to both arms: two such squares would have to overlap on both axes. Since the arms cover the four core squares, up to interchanging

axes there are just two possibilities: one arm contains three core squares, or $r = u = 2$ with disjoint core arms.

In Alternative I, a small square in $\mathcal{R}$ would be y-disjoint from all three core squares there, giving a chain excluded by (Q). Thus, all small squares lie in $\mathcal{U} \setminus \mathcal{R}$. The index k belongs to $\mathcal{U}$. Hence it is x-disjoint from all the small squares.

If $t \in \mathcal{U}$, the same is true of t, and k, t are x-disjoint. Otherwise $y_t < s_0$. Any small square, or Q_k, that x-overlapped Q_t would have to lie above it: its lower edge is at least $s_0 > y_t$. Together with the three right-arm core squares, it would make a chain excluded by (Q). Thus, Q_t is x-disjoint from Q_k and all three small squares.

In Alternative II, each core square belongs to exactly one arm. Hence $x_p, x_q < s_0 \le x_b, x_t$ and $y_b, y_t < s_0 \le y_p, y_q$. The disjoint squares Q_q, Q_t satisfy either "t right of q" or "q above t." Interchanging the axes exchanges the two arms, and hence exchanges the roles b, t with p, q, respectively; it turns the second alternative into the first. The square indices themselves are unchanged. Thus, p, q, t is a horizontal chain.

A small square in $\mathcal{R}$ x-overlaps both right-arm core squares, and hence is y-disjoint from both. A small square outside $\mathcal{R}$ belongs to $\mathcal{U}$; it is x-disjoint from p, q. It cannot also be x-disjoint from t, by (Q). It therefore x-overlaps t, is y-disjoint from it, and lies above it because its lower edge is at least $s_0 > y_t$. It also lies above b. This proves the assertions about all three small squares.

Under the additional exclusion in the lemma, two small squares cannot be y-disjoint, since together with b, t they would form a forbidden chain. Thus, the small squares y-overlap pairwise and are mutually x-disjoint. Order them from left to right as e_1, e_2, e_3. Since e_1 x-overlaps t, we have $x_{e_2} \ge x_{e_1} + s_{e_1} > x_t \ge x_q + s_q$. Thus, e_2, e_3 lie right of q, and also right of p. Finally, if one small square is outside $\mathcal{R}$, it lies above t by the preceding argument. Since all three are y-disjoint from t and are pairwise y-overlapping, all must lie on the same side of it, namely above. □

In either frame alternative, three core squares form a chain, and hence $L \ge s_2 + s_3 + s_4$: use the right-arm triple in Alternative I or p, q, t in Alternative II.

Proposition 4.9. *For every $s \in \ell_2^{\downarrow}$,*

$$L_8^*(s) \ge b_6(s) := \min(E,\ X),$$

where $X = s_3 + s_4 + s_6 + s_7$.

Proof. Put $\Sigma = b_6$ and suppose $L < \Sigma$.

Every triple in (C) has side sum at least E. Every four-square family containing at least two core indices has side sum at least X: for three or four core indices the additional core sides only increase this minimum. Thus, (C) and (Q) hold, and we may use the frame.

The frame lemma also gives $L \ge s_2 + s_3 + s_4$. Since $X > L$, it follows that $s_6 + s_7 > s_2 \ge s_4$. Consequently,

$$P_\chi - E = s_5 + s_6 - s_4 \ge s_6 + s_7 - s_4 > 0, \tag{4.6}$$

and $P_\chi > E \ge \Sigma$.

In Alternative I, any two x-disjoint small squares, together with k, t, would make a clearing four-square chain. Otherwise all three small squares are pairwise x-overlapping and are a vertical chain above Q_0. With Q_0, their side sum is P_χ, another contradiction.

In Alternative II, a chain consisting of b, t and two small squares has side sum at least X. The frame lemma therefore produces the horizontal chain consisting of p, q and the middle and rightmost small squares. This chain also has side sum at least X. Both alternatives are impossible. □

Proposition 4.10. *For every* $s \in \ell_2^{\downarrow}$,

$$L_8^*(s) \geq b_7(s) := \min(E,\ J_4,\ Y),$$

where $Y = s_2 + s_4 + s_6 + s_7$.

Proof. Put $\Sigma = b_7$ and suppose $L < \Sigma$. Call the core indices $1, 2$ marked. The following families clear Σ:

(1) the triples in (C), with side sum at least E;
(2) a triple of core indices containing both marked indices, with side sum at least J_4;
(3) four-square families having at least three core indices, with side sum at least Y;
(4) two core indices, at least one marked, and two small indices, with side sum at least Y.

For (3), the least sum with exactly three core indices is $s_2 + s_3 + s_4 + s_7 \geq Y$. For (4), the least sum is precisely Y. In particular (C) and (Q) hold.

Again the frame gives a three-core chain. Hence $L \geq s_2 + s_3 + s_4$, and $Y > L$ gives $s_6 + s_7 > s_3 \geq s_4$. It follows, as in (4.6), that $P_\chi > E \geq \Sigma$. Lemma 2.3 gives $L \geq s_0 + s_1$, and $E > L$ gives $s_3 + s_5 \geq s_4 + s_7 > s_1 \geq s_2$. Therefore, the five-square family with indices $3, 4, 5, 6, 7$ clears Σ:

$$s_3 + s_4 + s_5 + s_6 + s_7 = Y + (s_3 + s_5 - s_2) > Y \geq \Sigma. \tag{4.7}$$

In Alternative I, the right-arm triple cannot contain both marked indices. Thus, the outside core index k is marked. If two small squares are x-disjoint, they and k, t make a clearing chain of type (4). Otherwise the small squares and Q_0 form a vertical chain with side sum P_χ.

In Alternative II, the horizontal triple p, q, t cannot contain both marked indices. Hence b is marked. Since type (4) forbids a chain consisting of b, t and two small squares, the last conclusions of Lemma 4.8 apply. If either p or q is marked, the resulting chain with the middle and rightmost small squares is of type (4).

It remains that $\{p, q\} = \{3, 4\}$ and $\{b, t\} = \{1, 2\}$. If all small squares belong to $\mathcal{R}$, their horizontal chain together with Q_0 has side sum P_χ. Otherwise, by the last assertion of Lemma 4.8, all three lie above Q_t. Since Q_t lies above Q_b and $\{b, t\} = \{1, 2\}$, its top edge satisfies $y_t + s_t \geq s_1 + s_2 \geq s_2 + s_4 > s_0$, where the final inequality follows from $J_4 > L \geq s_0 + s_1$. Thus, all small squares belong to $\mathcal{U}$, and each is x-disjoint from p, q. With their mutual x-disjointness this gives the five-square horizontal chain $3, 4, 5, 6, 7$, which clears Σ by (4.7). □

Proposition 4.11. *For every $s \in \ell_2^{\downarrow}$,*

$$L_8^*(s) \geq b_8(s) := \min(E,\ H_4,\ M,\ P_\chi),$$

where $H_4 = s_1 + s_3 + s_4$ and $M = s_1 + s_4 + s_6 + s_7$.

Proof. Put $\Sigma = b_8$ and suppose $L < \Sigma$. The triples in (C) have side sum at least E. Every triple of core indices containing index 1 has side sum at least H_4. Since $E > L \geq s_0 + s_1$, we have $s_2 + s_7 \geq s_4 + s_7 > s_1$. Hence every three-core-plus-one-small family clears Σ, because $s_2 + s_3 + s_4 + s_7 = H_4 + (s_2 + s_7 - s_1) > H_4 \geq \Sigma$. The four-core family clears Σ as well. Thus, (C) and (Q) hold. Finally any family consisting of index 1, another core index, and two small indices has side sum at least M.

In Alternative I, the right-arm triple cannot contain index 1. Hence $k = 1$. Two x-disjoint small squares, together with k, t, would form a chain with side sum at least M. Otherwise the small squares and Q_0 form a vertical chain with side sum P_χ.

In Alternative II, the horizontal triple p, q, t cannot contain index 1. Hence $b = 1$. A chain consisting of b, t and two small squares has side sum at least M. Thus, the small squares y-overlap pairwise and form a horizontal chain. If all belong to $\mathcal{R}$, this chain with Q_0 has side sum P_χ.

Otherwise, by the last assertion of Lemma 4.8, all three lie above Q_t. The inequality $H_4 > L \geq s_0 + s_1$ implies $s_3 + s_4 > s_0$. Since Q_t lies above $Q_b = Q_1$, we have $y_t + s_t \geq s_1 + s_t \geq s_1 + s_4 \geq s_3 + s_4 > s_0$. All small squares therefore belong to $\mathcal{U}$, and are x-disjoint from p, q. The resulting five-square horizontal chain has sum $s_p + s_q + s_5 + s_6 + s_7 \geq s_3 + s_4 + s_5 + s_6 + s_7 > s_0 + s_5 + s_6 + s_7 = P_\chi \geq \Sigma$. This is the final contradiction. □

5. Proof of Theorem 1.1

5.1. **The lower bounds imply the formula.** We now use only the formulas (1.1) and (1.2) for L_6^* and L_7^* from [Bad26a, Theorems 3.8 and 5.5], the definition of F_8, the column bound, and the lower bounds $b_1, \ldots, b_8$. For $n = 8$, Lemma 2.4 gives $L_8^*(s) \geq \min(D, P)$ for all $s \in \ell_2^{\downarrow}$, where $D = s_1 + s_3 + s_7$ and $P = s_1 + s_4 + s_5 + s_6 + s_7$, the value of P_8. Using the notation of Table 4.1, the seven-square formula (1.2) reads

$$L_7^* = \max\bigl(L_6^*, \min(\max(W, H), \max(C_0, C_3), R)\bigr). \tag{5.1}$$

Table 5.1 collects the lower bounds from Section 4.

Table 5.1. The eight lower bounds of Section 4.

$b_1 = \min(T, \mathcal{K})$	$b_4 = \min(G_7, U, H_5, P_\pi, P_\rho, P_\chi)$	$b_7 = \min(E, J_4, Y)$
$b_2 = \min(C_0, E, J_4, J_{67})$	$b_5 = \min(K, G_6, U, H_5, J_6, V)$	$b_8 = \min(E, H_4, M, P_\chi)$
$b_3 = \min(J_7, V)$	$b_6 = \min(E, X)$	

Lemma 5.1 (Five balanced partitions)**.** *For every $s \in \ell_2^{\downarrow}$, the minimum defining $\mathcal{K}(s)$ in Proposition 4.1 is attained by one of the partitions*

	I	J
ξ	$\{1,3,7\}$	$\{2,4,5\}$
σ	$\{1,4,7\}$	$\{2,3,5\}$
τ	$\{1,4,5\}$	$\{2,3,7\}$
υ	$\{1,5,7\}$	$\{2,3,4\}$
φ	$\{1,2,7\}$	$\{3,4,5\}$.

The labels indicate the corresponding rows of Table 1.1. Each partition supplies two terms of that row after s_6 is added to its part-sums. These are all the terms of σ, τ, and υ; the rows ξ and φ also have the terms H_5 and H, respectively, treated in Lemma 5.2.

Proof. Identify each of the ten unordered $3+3$ partitions by its part containing 1. The φ partition has no larger maximum part-sum than any partition whose part containing 1 is $\{1,2,3\}$, $\{1,2,4\}$, or $\{1,2,5\}$. Indeed, both of its part-sums are at most $s_1+s_2+s_5$, which is at most the sum over each of these three parts. Similarly, the ξ partition has no larger maximum part-sum than any partition whose part containing 1 is $\{1,3,4\}$ or $\{1,3,5\}$, since both of its part-sums are at most $s_1+s_3+s_5$. These comparisons use only $s_0 \geq \cdots \geq s_7$. □

Lemma 5.2 (Truncation at T)**.** *If $F_8 \leq T$, then* $\max(L_7^*, b_1) \geq F_8$.

Proof. If $L_7^* \geq T$, then $F_8 = L_7^* = T$. Suppose $L_7^* < T$. Since $L_7^* \geq s_0 + s_1$, we have $s_1 < s_6+s_7$. The additional terms H_5 in ξ and H in φ are redundant: $H_5 \leq s_2+s_4+s_5+s_6$ and $H \leq J_{67}$. For the first comparison use $s_2 \geq s_3$ and $s_4+s_6 \geq s_6+s_7 > s_1$. For the second use $s_5 \leq s_1 < s_6+s_7$. Thus, by Lemma 5.1, $\min(\xi,\sigma,\tau,\upsilon,\varphi) = \mathcal{K}$.

Every remaining maximum except ψ has a term at least T. In Table 1.1, these terms are C_0 for $\alpha,\beta,\gamma,\varepsilon,\theta,\iota,\kappa,\mu,\pi,\rho$; U for δ,λ; G_6 for η; T for ζ,ν; E for o; and P_χ for χ. Here $\psi = P$ is no smaller than τ. Indeed $R < P$, while $P - (s_2+s_3+s_6+s_7) = (s_1-s_2)+(s_4+s_5-s_3) > 0$ because $s_4+s_5 \geq s_6+s_7 > s_1 \geq s_3$. Consequently, $\min(T, \min_\omega \omega) = \min(T, \mathcal{K})$. Since $F_8 = \max(L_7^*, \min_\omega \omega)$ and $L_7^* < T$, it follows that

$$\min(F_8, T) = \max\Big(L_7^*, \min\Big(T, \min_{\omega} \omega\Big)\Big) = \max(L_7^*, b_1). \tag{5.2}$$

The left-hand side equals F_8 because $F_8 \leq T$. □

Theorem 5.3. *For every ordered positive eight-tuple,*

$$\mathcal{L}(s) := \max\{L_7^*, \min(D,P), b_1, \ldots, b_8\} \geq F_8(s).$$

Proof. Suppose instead that $\mathcal{L} < F_8$. Then $L_7^* < F_8$. Hence F_8 equals the minimum of the side-sum maxima and every such maximum is at least F_8. In particular, $P = \psi \geq F_8$.

We first use the column bound and the bounds b_1 and b_2. Lemma 5.2 gives $T < F_8$. Since $\min(D,P) < F_8$, we also have $D < F_8$. Comparing with $\nu = \max(T,D,R)$ gives

$R \geq F_8$. Moreover $C_3 < F_8$; indeed $H \geq C_3$ because $s_2 \geq s_3$ and $s_5 \geq s_6$, and (5.1) gives $L_7^* \geq \min(C_3, R)$. Since $L_7^* < F_8 \leq R$, this gives $C_3 < F_8$.

If $C_0 \geq F_8$, then the entries $\max(C_0, C_3)$ and R of the minimum in (5.1) are at least F_8. Thus, $L_7^* < F_8$ forces $W, H < F_8$. Comparison with $\zeta = \max(W, J_4, T)$, $o = \max(W, E, H)$, and $\varphi = \max(W, J_{67}, H)$ gives $J_4, E, J_{67} \geq F_8$, respectively. Then $b_2 \geq F_8$, a contradiction. Consequently, $\max(L_7^*, C_0, C_3) < F_8 \leq R$. The three maxima π, ρ, χ are, respectively, $\max(C_0, P_\pi)$, $\max(C_0, C_3, P_\rho)$, and $\max(C_3, P_\chi)$. Since each is at least F_8, we conclude that

$$P_\pi, P_\rho, P_\chi \geq F_8. \tag{5.3}$$

Next we use the bounds b_3, b_4, and b_5. Suppose $H_5 \geq F_8$. If $U < F_8$, comparison with $\delta = \max(U, J_7, C_3)$ and $\lambda = \max(V, C_3, U)$ gives $J_7, V \geq F_8$, contradicting $b_3 < F_8$. Thus, $U \geq F_8$. If $G_7 \geq F_8$, then (5.3) gives $b_4 \geq F_8$, also impossible. Hence $G_7 < F_8$. Comparison with $\beta = \max(V, C_0, G_7, C_3)$ and $\gamma = \max(C_0, G_7, J_6)$ gives $V, J_6 \geq F_8$. If $J_7 \geq F_8$, then $b_3 \geq F_8$, again a contradiction. Otherwise $J_7 < F_8$, and the inequality $\eta = \max(G_6, J_7) \geq F_8$ gives $G_6 \geq F_8$. Since $C_0, C_3 < F_8$, the inequality $\theta = \max(K, C_0, C_3) \geq F_8$ gives $K \geq F_8$. Now $b_5 \geq F_8$, a contradiction. Therefore, $H_5 < F_8$.

Finally, we use the bounds b_6, b_7, and b_8. If $E < F_8$, then comparison with $\alpha = \max(J_7, C_0, E, H_5)$ and $\iota = \max(V, C_0, H_5, E)$ gives $J_7 \geq F_8$ and $V \geq F_8$, contradicting $b_3 < F_8$. Thus, $\max(L_7^*, C_0, H_5) < F_8 \leq E$. The remaining comparisons are

$$\kappa = \max(C_0, J_4, X), \qquad \varepsilon = \max(C_0, H_4, Y),$$
$$\mu = \max(C_0, H_5, M), \qquad \chi = \max(C_3, P_\chi).$$

Hence $\max(J_4, X) \geq F_8$, $\max(H_4, Y) \geq F_8$, and $M, P_\chi \geq F_8$. If $X \geq F_8$, then $b_6 \geq F_8$. Otherwise $J_4 \geq F_8$. If $Y \geq F_8$, then $b_7 \geq F_8$. Otherwise $H_4 \geq F_8$, and $b_8 \geq F_8$. Each alternative contradicts $\mathcal{L} < F_8$. □

Proof of the formula in Theorem 1.1. Fix $s \in \ell_2^{\downarrow}$. Deletion of Q_7 gives $L_8^* \geq L_7^*$, Lemma 2.4 gives $L_8^* \geq \min(D, P)$, and Section 4 gives $L_8^* \geq b_i$ for $i = 1, \ldots, 8$. Since L_7^*, $\min(D, P)$, $b_1, \ldots, b_8$, and F_8 are continuous, Theorem 5.3 extends from positive tuples to all of $\ell_2^{\downarrow}$. Therefore, $L_8^* \geq \mathcal{L} \geq F_8$. Proposition 3.1 gives the reverse inequality $L_8^* \leq \min_\omega \ell_\omega = F_8$, which proves both identities. □

5.2. **Minimality of convex representations.** Every strategy cost ℓ_σ is convex and positively homogeneous. In particular it is subadditive: $\ell_\sigma(s + t) \leq \ell_\sigma(s) + \ell_\sigma(t)$. The minimum $L_n^*(s)$ need not be subadditive in s. A strict reverse inequality $L_n^*(s + t) > L_n^*(s) + L_n^*(t)$ shows that no single strategy is optimal at both s and t: if $\ell_\sigma(s) = L_n^*(s)$ and $\ell_\sigma(t) = L_n^*(t)$, then $L_n^*(s + t) \leq \ell_\sigma(s + t) \leq L_n^*(s) + L_n^*(t)$.

Lemma 5.4 (Counting convex costs, [Bad26a, Lemma 6.1])**.** *Let D be a convex cone and $F : D \to \mathbb{R}$ positively homogeneous. Suppose $p_1, \ldots, p_N \in D$ satisfy*

$$F(p_i + p_j) > F(p_i) + F(p_j) \quad \text{for all } i < j. \tag{5.4}$$

Every representation $F = \min_{1 \leq k \leq m} h_k$ with convex h_k on D requires $m \geq N$.

ω	p_ω	$L_8^*(p_\omega)$	ω	p_ω	$L_8^*(p_\omega)$
α	$(21, 18, 15, 13, 12, 10, 9, 8)$	41	ν	$(195, 194, 162, 161, 66, 65, 64, 6)$	389
β	$(21, 19, 16, 13, 11, 10, 8, 7)$	41	ξ	$(71, 32, 31, 27, 26, 24, 23, 22)$	104
γ	$(20, 18, 15, 14, 12, 11, 9, 8)$	42	o	$(225, 166, 128, 100, 98, 97, 96, 68)$	391
δ	$(85, 83, 69, 67, 53, 52, 39, 38)$	190	π	$(39, 27, 24, 23, 19, 18, 9, 8)$	67
ε	$(61, 38, 33, 32, 31, 20, 19, 18)$	101	ρ	$(38, 29, 26, 25, 16, 15, 11, 8)$	68
ζ	$(690, 413, 412, 279, 278, 277, 269, 144)$	1103	σ	$(73, 34, 31, 28, 27, 25, 24, 23)$	108
η	$(164, 162, 131, 128, 98, 97, 66, 65)$	358	τ	$(527, 212, 211, 181, 177, 176, 175, 173)$	740
θ	$(71, 69, 54, 52, 50, 37, 35, 32)$	157	υ	$(77, 38, 31, 30, 29, 27, 26, 25)$	116
ι	$(50, 47, 38, 30, 29, 21, 20, 19)$	98	φ	$(1034, 466, 433, 369, 368, 367, 366, 236)$	1501
κ	$(952, 580, 579, 478, 477, 342, 341, 340)$	1636	χ	$(98, 87, 66, 65, 54, 35, 27, 26)$	186
λ	$(26, 23, 20, 17, 13, 12, 10, 9)$	51	ψ	$(265, 264, 231, 230, 68, 67, 66, 65)$	530
μ	$(38, 27, 23, 22, 20, 17, 10, 9)$	66			

TABLE 5.2. Ordered tuples p_ω with $L_8^*(p_\omega + p_{\omega'}) > L_8^*(p_\omega) + L_8^*(p_{\omega'})$ for all $\omega \neq \omega'$. At p_ω, the strategy σ_ω is the only one of the 23 with cost $L_8^*(p_\omega)$.

Proof of the bound $m \geq 23$ in Theorem 1.1. The twenty-three strategies of Theorem 1.1 form a complete family, and their costs are convex. For the lower bound, use the twenty-three integer vectors in Table 5.2. Substitution into the formula of Theorem 1.1 at those vectors and their 253 pairwise sums gives

$$L_8^*(p_\omega + p_{\omega'}) > L_8^*(p_\omega) + L_8^*(p_{\omega'}) \quad \text{for all } \omega \neq \omega'. \tag{5.5}$$

For instance, Table 5.2 gives $L_8^*(p_\alpha) = L_8^*(p_\beta) = 41$, while direct substitution yields $L_8^*(p_\alpha + p_\beta) = 83 > 82$. All 253 differences are positive integers. These are finite integer computations with the formula of Theorem 1.1. The accompanying code [Bad26b] verifies these computations and checks its data against the manuscript tables. Lemma 5.4 requires at least twenty-three convex functions, hence at least twenty-three strategies. □

Corollary 5.5 (Unit-container parameter sets)**.** *The set $X_8 = \{s \in \ell_2^\downarrow : L_8^*(s) \leq 1\}$ is a union of twenty-three convex subsets and not of fewer.*

Proof. By Theorem 1.1, X_8 is the union of the sublevel sets $\{\ell_\omega \leq 1\}$ of the twenty-three convex strategy costs. For the last assertion, put $\hat{p}_\omega = p_\omega / L_8^*(p_\omega) \in X_8$ for the vectors p_ω in Table 5.2. For $\omega \neq \omega'$, the point $(p_\omega + p_{\omega'})/(L_8^*(p_\omega) + L_8^*(p_{\omega'}))$ lies on the segment joining $\hat{p}_\omega$ and $\hat{p}_{\omega'}$, and by (5.5) and positive homogeneity its optimal value is greater than one. Thus, no convex subset of X_8 contains two of the points $\hat{p}_\omega$, whereas one of fewer than twenty-three sets covering X_8 would contain two of them. □

DEVELOPMENT OF THE RESULTS AND AI ASSISTANCE

I found the candidate formula for L_8^* in the second half of July 2026, using Claude Opus 4.8 (Anthropic) as a research assistant. The search was guided by the form $L_n^* =$

$\max(L^*_{n-1}, \cdot)$ of the formulas for $n \le 7$. The AI system computed exact optima for integer side vectors with the constraint-programming solver CP-SAT of Google OR-Tools, and compared them with the minimum of the costs of the strategies found so far. A side vector with $L^*_7 < L^*_8$ at which this minimum exceeds L^*_8 shows that a strategy is missing. I first restricted the searches to the region $L^*_8 < s_4 + s_5 + s_6 + s_7$, where no four squares can form a chain, so that every chain determining the container side has at most three squares. They produced such side vectors, and new strategies were read off their optimal packings until exhaustive searches over integer side vectors with $s_0 \le 40$ found none there.

Outside that region, I inserted the eighth square into the three seven-square arrangements of [Bad26a] at geometrically reasonable places and read candidate terms off the chains that determine the container side. This gave six further strategies. The search then became a loop. I chose pairs of side vectors at which the current strategies failed, the AI system computed and drew their optimal packings, and I proposed a single arrangement serving both. The AI system then checked the proposal against exact optima at all nonincreasing integer side vectors with $s_0 \le 17$ and at random side vectors, and occasionally repaired a term. Extracting the separation patterns and chain sums from the optimal packings required my intervention; on one occasion the AI system read a term off the arithmetic rather than the geometry, and I read the correct term off the picture. To normalize the scale, I ran the later searches on slices with s_0 fixed, and a strategy was added only after a side vector was found at which it is needed. I found the last strategy, σ_ψ, associated with the five-term sum $s_1 + s_4 + s_5 + s_6 + s_7$, by scaling and perturbing a side vector with four large and four small squares.

By the end of July, an exhaustive search over strictly decreasing positive integer side vectors with $s_0 = 44$ (about 32 million side vectors and 450,000 exact solves) found no side vector at which the twenty-three strategies fail to attain L^*_8, and each strategy was the only one attaining L^*_8 at some side vector there. The Greek letters order the strategies by the number of such side vectors, from α (most) to ψ (fewest).

Beginning in late July, I attempted the lower bound one strategy at a time, starting with α. With the assistance of Claude Opus 4.8, I established the desired lower bound on two of the regions determined by which term of α is largest, using a cornering argument for the largest square. The remaining regions resisted this approach, and I paused work on square packings in early August to work on other projects. On October 7, 2026, I supplied GPT-6 Astra Pro (OpenAI) with the candidate formula and the seven-square proof and asked it to develop lower bounds using those methods. Over a sequence of exchanges, it developed and assembled proofs of the eight new inequalities and their arithmetic combination that, after revision, are presented in Sections 4 and 5. Subsequent exchanges led to the convex-minimality argument and its supporting certificates.

The manuscript was written and revised using Claude Opus 5.5 (Anthropic) and GPT-6 Astra Pro, which also reviewed the mathematical arguments. Claude Opus 5.5 prepared

the verification code [Bad26b] and used it to check the constructions, figures, and certificates in exact arithmetic. I have checked all proofs and am responsible for the content of this paper.

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